

UNIVERSITÄT PADERBORN
Die Universität der Informationsgesellschaft

Modelling and Forecasting Seasonal Time Series using the ARMA-version of the Berlin Method

Master's Thesis in Economics

Author: Jing Ma

Student ID: 6403914

Email: majing925@hotmail.com

Supervisor: Prof. Dr. Yuanhua Feng

November 5, 2009

Faculty of Business Administration and Economics

University of Paderborn

Acknowledgements

I would like to take this opportunity to express my deepest gratitude to my supervisor Prof. Dr. Yuanhua Feng, whose helpful suggestions have contributed greatly to the completion of this thesis. Thanks for his insightful lectures, which inspire me to compose this thesis, and for his allowing me to have access to his books pertinent to this thesis. Without his consistent and effective instructions, constant encouragement and support throughout the course of the research, this thesis could not have reached its present form. In addition, the data for the time series used in application are downloaded from the *Time Series Data Library*. I would like to thank Prof.Rob J. Hyndman, Monash University, for making these data publicly available.

Abstract

This paper presents a theoretical introduction of nonparametric techniques and discusses decomposition of seasonal time series based on locally weighted regression. There are a large number of studies and proposals have showed that nonparametric techniques can solve the misspecification problem of parametric techniques. Two widely used methods-kernel regression and local regression involving the asymptotic properties are investigated in great detail.

Decomposing time series into trend-cyclical, seasonal components has a long tradition. Since 1960s quite a large number methods and software systems are developed to analyze this. In this paper an extension of the fourth version of the Berlin-Method (BV4, Berliner Verfahren 4) is used , which is a seasonal adjustment method based on moving filter applications derived from approximating functions by nonparametric regression approach. The basic model is the traditional additive model for economic time series with a trend component, a seasonal component and a stationary residual process. The first two components are assumed to be deterministic and can be modeled by nonparametric techniques, in particular using locally weighted regression. Moreover, an improvement of this paper is to model the residual term by a stationary autoregressive and moving average (ARMA) process. Another focus of this paper is the short-term forecasting of seasonal time series based on fitted model.

A crucial problem for nonparametric regression and time series decomposition is the data-driven selection of bandwidth. It is important to make a trade-off between modeling bias and variance. Some popular selection methods, e.g. Cross-validation (CV), the R statistic and double smoothing (DS) method are briefly investigated. Here a new algorithm based on iterative plug-in method will be proposed in great detail.

Keywords: Berlin Method; Kernel Regression; Local Polynomial Regression; ARMA Model; Seasonal Adjustment; Plug-In

Contents

Acknowledgements ..... *I*

Abstract ..... *II*

Contents ..... *III*

1. Introduction *1*

2. Literature Review *4*

3. Semiparametric Regression Model with ARMA Errors without Seasonality *10*

3.1. Introduction **10**

3.2. Kernel Regression **13**

3.2.1. Kernel Estimators 13

3.2.2. Kernel Functions 17

3.2.3. Asymptotic Properties 20

3.3. Local Polynomial Regression **24**

3.3.1. Advantages of Local Polynomial Fitting 24

3.3.2. Local Polynomial Estimators 24

3.3.3. Finite Sample Properties 27

3.3.4. Asymptotic Properties 28

3.4. Estimation of ARMA Model **31**

3.4.1. Estimation of the AR Model 31

3.4.2. Estimation of the ARMA Model 34

3.5. Forecasting for ARMA Model 36

3.5.1. Point Forecasting for AR Model 36

3.5.2. Interval Forecasting for ARMA Model 38

4. Semiparametric Regression Model with ARMA Errors and Seasonality 40

4.1. Time Series Decomposition with LWR 40

4.2. Finite Sample Properties 44

4.3. Asymptotic Properties 45

5. Bandwidth Selection 48

5.1. Introduction 48

5.1.1. Error Criteria 49

5.1.2. The R Criterion 50

5.1.3. Cross-Validation 50

5.1.4. Double Smoothing 51

5.2. Plug-In 52

5.2.1. Estimation of Variance 53

5.2.2. Estimation of I 55

5.2.3. The Basic Algorithm 56

6. Applications 59

6.1. Example 1 59

6.2. Example 2 66

6.3. Example 3 72

7. Concluding Remarks 78

Appendix 80

References 85

Eidesstattliche Erklärung 90

1. Introduction

In the real economic phenomena, many economic time series have trend-cycle component and seasonal component due to e.g. business cycle, economic development and natural or calendar periods. The main purpose of this paper is to analyze some real economic seasonal time series of national GDP and some important macroeconomic indices by using nonparametric techniques and autoregressive and moving average (ARMA) model. In addition, since forecasting is an important objective of many time series analyses, another focus of this paper is the short-term forecasting based on the fitted model. The basic model is additively composed by a trend component for the business cycle, a seasonal component for yearly periodic change and a stationary residual process. Trend and seasonality can be modeled by locally weighted regression, while the residual term will be modeled by an ARMA process. So the whole model is said to be a semiparametric time series model.

Decomposing time series into trend-cyclical, seasonal components has a long tradition. Many different methodical seasonal adjustment approaches and associated software systems have been developed over time. A complete historical review can be found in Hylleberg (1986). The study of Fisher (1994) showed a methodical and simulated comparison between the related methods. Heiler (1995) provided a survey of methods in this field from the begging of the 1960s up to the newest developments and proposals. Contributions to this field also can be found in Eurostat (1998). Section 2 provides a short description and theoretical judgment of the methods in this area including the Census X-11 Method, the X-12 ARIMA, the Dainties, and the SEATS. In particular we will focus on the BV method, where BV stands for “Berliner Verfahren”.

Nonparametric and semiparametric methods have attracted a great deal of attentions. Section 3 will present the basic concepts and related results of nonparametric regression, while the local regression applied to time series decomposition will be discussed in the next section.

Compared with parametric regression with specified assumptions, the aim of nonparametric techniques is to relax the restrictive conditions from a regression function; it provides a better tool for investigating an object estimated. In practice most of the models in economic time series are often very complex, so we cannot make sure, what for a parametric model should be used, and whether or not it is suitable. The use of an inaccurate parametric model will lead to the bias of the resulting estimators and consequently influences the residuals and the fitted model. Hence, in order to solve the misspecification of the parametric model, the nonparametric model should be applied. There exist several methods for obtaining nonparametric regression estimator, two of the most commonly used approaches are smoothing splines and kernel regression. In this section we will give emphasis on kernel regression and local polynomial regression. Kernel smoothing, which has the advantage of being very intuitive and relatively simple to analyze mathematically, is the most widely used nonparametric method. At the outset, we offer an analytical presentation and evaluation of kernel regression, which was introduced in 1964 by Nadaraya and Watson independently, and then followed by the description of the local polynomial regression. Local polynomial fitting is actually an extension of kernel regression, so these two approaches have some similar sample properties, which are the basis for applying a plug-in bandwidth selection method in section 5.

Additionally, in this section we also give a short introduction of ARMA model. The notation ARMA (p, q) refers to the model with p autoregressive terms and q moving average terms. Further, estimation and forecasting of ARMA model will be discussed. Besides, two criterions for choosing suitable parameters - BIC (Bayesian's information criterion) and AIC (Akaike's information criterion) will be introduced. After that, we give the calculation process for the point forecasting of AR model and interval forecasting of ARMA model.

Section 4 discusses the seasonal time series decomposition by using the BV method based on the locally weighted regression. This section is mainly based on the proposal of Heiler and Michels (1994). Also, its finite sample properties and asymptotic properties will be

investigated, not only for the independent and identical distribution case, but also that in the case of ARMA error.

One important issue of nonparametric regression is the choice of the right amount of smoothing, called the bandwidth or smoothing parameter denoted by h. Nonparametric regression performs well, only if the bandwidth h is suitable. A very small h will lead to a small modeling bias; however, the variance will be large. On the other hand, if the bandwidth h is very large, it can create a large modeling bias, whereas the variance will be small. Hence, the selection of appropriate bandwidth is of great importance. And we must make a trade-off between modeling bias and variance. Section 5 will deal with this smoothing parameter selection problem in great detail. First, we investigate some popular bandwidth selection rules, such as cross-validation (CV), R statistic, error criterion, and double smoothing (DS) method. Then we will focus on the plug-in method. A seasonal-difference-based variance estimator in a seasonal time series was proposed by Heiler and Feng (1996), as well as the basic algorithm will be presented.

Section 6 is concerned with practical performance with some data examples. Data used in this paper were mainly obtained from *Time Series Data Library* (http://robjhyndman.com/TSDL). And the data includes among consumption, gross national product, external trade and money supply. Besides, in order to carry out the estimation problem in time series analysis, we will use the R project for statistic computing and graphics, which is the best free software with powerful time series analysis function. Finally, in section 7, we will give some concluding remarks.

2. Literature Review

There are a large number of different methodical approaches and software systems for the decomposition and seasonal adjustment of monthly and quarterly economic time series since 1960s, e.g. the Census X-11 Method (Eisenpress 1956 , Marris 1960 and Young 1965), the Berlin Method (BV, Heiler, 1970), the SABL (Seasonal Adjustment at the Bell Laboratories ,Cleveland, Dunn and Terpenning, 1978), the Dainties (1979), the X-12 ARIMA (Dagum,1988), and the SEATS (Signal Extraction in ARIMA Time Series, Maravall and Gomez, 1994). These procedures are derived from different mathematical-statistical techniques which may thus lead to differing results. First, we provide a brief introduction of the above most commonly used methods with a theoretical judgment of the advantages and disadvantages, and then we will concentrate on the Berlin method.

As described in Eurostat (1998), the Census X-11 Method, which is still the most attacked method for seasonal adjustment based on different kinds of moving average methods, was developed in 1965 by Julius Shiskin, who has made a major contribution to the development of this method. It can handle additive models and log-additive models. Its first version-Census I method was developed by the U.S. Bureau of the Census in 1954, and later it was revised to create Census II method in 1955. The main advantage of X-11 Method is its robust adjustment because of the outlier treatment, and it is developed from practitioners for the practical purposes such as trading day, outlier, moving holidays. However, it is just an ad hoc method，the measurements are therefore usually unreliable. A variant of X-11 is the X-11 ARIMA developed by Dagum and Laniel in 1987. It had an improvement at dealing with missing data compared with X-11 method, since the X-11 extrapolated the missing values arbitrarily. In contrast, as presented in Eurostat (1998, p.4), “It used an ARIMA (Auto Regressive Integrated Moving Average) model to forecast beyond the current series and backcast before the beginning of the series in order to replace the missing data which results at the beginning and end of the series to allow the use of less asymmetric filters.”

The next development of X-11 family is the X-12 ARIMA, which was developed by David Findley, Brian Monsell, William Bell and others. From 1998 on, X-12 ARIMA became the official seasonal adjustment program of the US Bureau of the Census. Shortly after (2001), as a special service for the most important economic indicators, the Federal Statistical Office released seasonally adjusted series based on the X-12-ARIMA procedure besides the analysis results in accordance with the BV4.1 procedure. These results were identified in coordination with the German Federal Bank.

The program consists of three parts. The first part is Reg-ARIMA modeling of a time series. The use of the Reg-ARIMA model, which is a regression model with ARIMA errors, is effective for improving the procedure of seasonal adjustment, since it is possible for users to use an entirely new range of outlier and calendar adjustment capabilities. Besides, it is possible to extend the model through forecasts and backcasts.

Speth, H (2004) argued that, the central part of the procedure is identical to X-11 procedure. It means that the basis of seasonal adjustment is the iterative employment of several moving averages. The users must decide that how the time series components are composed, additively or multiplicatively.

The third part of this procedure is diagnostics. To obtain high-quality results, X-12 ARIMA requires the user to make a reasonable choice of model specification from a number of options. For this purpose, this part provides a large set of diagnostic instruments such as AIC, BIC, ACF, PACF, Ljung-Box test, which are helpful for the user to verify that whether the options are adequately selected, and to evaluate the results. Other advantages are showed as follows: discover and calculate addition components; estimate missing data; detect and adjust for outliers and other misrepresenting effects to improve the forecast and the seasonal adjustment; select between competing previous adjustments.

The SABL (Seasonal Adjustment at the Bell Laboratories) was developed at the Bell Laboratories in 1978. It could be an additive model, a multiplicative model based on log-

additive transformation or all kind of power transformation of time series. According to Eurostat (1998, p.9), “The method is based on a preliminary transformation of the series and in subsequent recursive smoothing to compute the seasonally adjusted series.” Moreover, the smoothing of robust nonlinear filters is followed by moving averages. The advantages of SABL are the robust estimation of the component and the power transformation of the data. However, SABL has not been improved and updated compared with X-11, which has improved with the ARIMA forecasting of Statistics Canada.

The Dainties, which was developed in 1979, was the official seasonal adjustment method of the European Commission. This method is based on regression approach over moving intervals, which is similar to the Berliner Method. The possible decomposition model could be additive, multiplicative based on log-additive transformation of time series, or mixed. The advantage of the Dainties is that, first, it is simple to calculate; second, it uses only asymmetric filter so that no revisions as new data are produced. Meanwhile, it suffers from some drawbacks. It cannot discover the seasonality changing. Another disadvantage is that it is impossible to illustrate a trend figure.

Unlike other methods, the SEATS (Signal Extraction in ARIMA Time Series) developed by Maravall and Gomez in 1994 is a parametric approach, which is based on a seasonal ARIMA model. SEATS offers not only many analytical tools like estimator errors and forecast errors, but also several good and complete graphical tools like graphs for the autocorrelation function (acf) of the residuals. The main advantages of this method are the basis of statistical inference and the adaptive filters to the characteristics of the time series. However, the validity of these advantages should meet requirements, that is, to model the time series adequately. Other description and analysis of this method can be found in Stier (1996), Planas (1997), Gomez and Maravall (1998).

The mathematical bases of Berlin Method (BV), which is applicable to analysis of monthly and quarterly economic time series, were developed originally by the Berlin Technical University and the German Institute for Economic Research (DIW) in the late 1960s. In 1972,

the Federal Statistical Office constructed a first practicable version of the procedure, which is convenient for the general public to search and use the information on trends and seasonally adjusted data of main business-cycle indicators. It has been applied for more than 30 years. In 1994, Prof. Heiler and Michels developed this method based on a traditional nonparametric technique, local regression with polynomials and trigonometric functions as local regressors, which makes it possible to estimate the trend-cyclical and the seasonal components in a unified approach. Its fourth version (BV 4), a non-data-driven high quality procedure, was developed by Nourney in 1983 and still in use until 2004, since it was replaced by version 4.1.

The method assumes a series to be additively composed by four components, namely trend-cycle, seasonal, calendar and residual or irregular fluctuations. The trend-cycle fluctuations relate to the impact of the factors reflecting the medium to long-term basic tendency, such as population growth, technological process, and political actions. The seasonal fluctuations comprise the impact of factors having the influences at annual interval such as climatic conditions and institutional events. What is meant by calendar fluctuations is the influences exerted on time series observations, for example, different weekday structures per month or quarter, various holidays, plant holidays. The residual or irregular fluctuations represent the effects of all other unpredictable factors that having the short-term influences, which may be caused by strikes, errors in data collections or natural disasters.

According to the proposal in Eurostat (1998), the trend-cycle component G_t (t = 1, ..., m) is approximated by a polynomial of order 3, while the seasonal component S_t is approximated by 11 trigonometric functions for monthly series:

$$\hat{G}_t = \sum_{i=0}^{3} \hat{a}_i t^i$$

$$\hat{S}_t = \sum_{i=1}^{6} \hat{b}_i \cos\lambda_i t + \sum_{i=1}^{5} \hat{c}_i \sin\lambda_i t$$

“The parameters are estimated by the weighted least squares. The estimation is performed not in the whole time series but in partial time intervals of different lengths moving over the whole time series -called reference intervals. In the middle of the time series for the seasonal component an interval length of 47 and for the trend-cycle component an interval length of 27 is chosen. The estimation point within the interval is chosen as t = 14 for the trend-cycle component and t = 24 for the seasonal component, so that a symmetric filter is used. At both ends of the time series, several asymmetric filters are used. The appropriateness of the symmetric and asymmetric filters is judged by comparing their transfer functions to "optimal" transfer functions.” Eurostat (1998, p.77)

The advantages of BV4 are summarized as following:

- The analysis results are easy to be understood and the calculation is quite quickly.
- There is generally no difference between the results of direct (i.e. the seasonal adjustment is performed on gross aggregates, computed using gross primary series) and indirect (i.e. aggregation of seasonal adjusted figures for seasonal adjusted aggregates) analyses, since the model is additively composed.
- Applying the program does not require any user training, long experience or expert knowledge.

Finally, Table 2.1 offers a summary of five common procedures.

Table 2.1: Summary of the methods

Method	Dainties	X-11 ARIMA	SABL	BV 4	SEATS
Type of approach	regression approach over moving intervals	Moving average type	Moving robust nonlinear filters followed by moving averages	regression method over a moving interval	model based approach
Possible decomposition	Additive, log additive, mixed	Additive, log additive, multiplicative	additive, log-additive, all kind of power transformation	additive	additive, log additive
Seasonal component	11 harmonic functions	smoothing with moving averages over monthly subseries	smoothing monthly subseries with moving averages after applying robust filters on the series	11 harmonics	extraction out of pseudo spectrum from estimated ARIMA model of the original series
Trend component	only as residuals because of direct estimation of seasonally adjusted series	9,13, 23 term Henderson filter to the seasonally adjusted series with outliers removed	moving averages after applying robust filters	cubic function over fixed intervals	Same as Seasonal component
Periodically	quarterly, monthly, half yearly	monthly, quarterly	any integer higher than 1	monthly, in UNIX version monthly and quarterly	any integer higher than one
Main advantage	no revisions	widespread use therefore currently updated	robust estimators	comparability of results, speed	adaptive to the characteristics of the time series, basis for statistical inference

Source: Seasonal Adjustment Methods: A Comparison, Eurostat, 1998, p. 36

3. Semiparametric Regression Model with ARMA Errors without Seasonality

3.1. Introduction

Assume we have a data set of variable X and Y $\{(x_i, y_i), i=1, \ldots, n\}$.Considered that these two variables have an approximate functional relationship, or more specific the two variables have the relationship defined by $Y_i = m(x_i) + \varepsilon_i$, where ε is the error term. How to estimate the functional form m(x) is our main task. For this problem, in general there are two methods. One is the parametric techniques, i.e. m(x) is some known, smooth function, and can be parameterized in terms of basic functions. In order to obtain the estimator of m(x), we only need to solve the estimation of a finite number of parameters. CLPM and non-linear regressions are types of this method. Parametric method has many advantages, especially the expression is simple and intuitive, and the employment is convenient. However, the real world is very complex, not all cases can be expressed by a finite mathematical relationship. Compare with the parametric technique, which has restrictive assumptions like smoothness, the nonparametric smoothing approach provides a flexible tool in analyzing unknown regression relationships. Here m(x) is some unknown, smooth function and unspecified in advance. Obviously, due to the existence of the error term, the data curve is not smooth. One simple method is moving average, which is used to smooth out short-term fluctuations and to make the long-term trends or cycles significant in time series. The each value of m(x) is taken from the average value of the data points, which close to x. Generally the more points close to x are used to be averaged, the smoother the curve is. If we use all n data points, m(x) will be constant and the smoothest. However, it will lose all useful information except mean value. So the problem is that, not only to remove the error as much as possible, but also to prevent the useful information from losing. Furthermore, there exists another problem, how to average, i.e. the role of each data point to estimate the value of m(x). It is clear that, not every point has the same function or weight. Intuitively, the points closest to x receive more weight than those

more remote from x, and often, points remote from x receive little or no weight. Therefore, the key point is the way of weighted average for smoothing. The methods that will be discussed later in this section can be collectively referred to as smoothing.

In this area there exist variety nonparametric techniques. However in this section, we will study some basic concepts of two most well known methods- kernel regression and local polynomial regression. From now on, we only consider the fixed design case. The basic nonparametric regression model of this section is given by

$$Y_t = m(x_t) + Z_t, \qquad t = 1, 2, \dots, n, \tag{3.1}$$

where Y_t are measurements of m at x_t , the design points x_t are on a compact set on R, for the sake of generality we assume $x_t \in [0,1]$, $0 \le x_1 \le \cdots \le x_n \le 1$. For equidistant time series x_t is the rescaled time $x_t = \frac{t-0.5}{n}$. And m is an unknown smooth function, Z_t are the residual errors. The conditional expectation m(x) of Y relative to X can be written as $E(Y \mid X=x) = m(x)$. The aim of smoothing techniques is to find suitable approximation $\hat{m}$ of the unknown function m. Note that if Y_t are independent, then Z_t are assumed to be independently and identically distributed (i.i.d.) random variables with zero-mean and a finite variance σ^2. In contrast, Y_t are dependent, and then Z_t are proposed to be a stationary Autoregressive Moving Average (ARMA) process. The ARMA process can be transformed to be i.i.d, it means that $Z_t = \epsilon_t$, $\sigma_z^2 = \sigma^2$. Next we give a short introduction of the ARMA model.

Given a time series Z_t, the ARMA model is used as a tool to understand and predict reasonable future values in the time series. An ARMA process of order (p, q) is defined by the following function, which consists of two parts, an autoregressive (AR) model and a moving average (MA) model.

$$Z_t = \beta_1 Z_{t-1} + \cdots + \beta_p Z_{t-p} + \alpha_1 \epsilon_{t-1} + \cdots + \alpha_q \epsilon_{t-q} + \epsilon_t \tag{3.2}$$

$$\Longrightarrow \quad Z_t - \beta_1 Z_{t-1} - \cdots - \beta_p Z_{t-p} = \alpha_1 \epsilon_{t-1} + \cdots + \alpha_q \epsilon_{t-q} + \epsilon_t$$

$$\Longrightarrow \quad \phi(B)Z_t = \psi(B)\epsilon_t \tag{3.3}$$

Where $$\phi(B) = 1 - \beta_1 B - \cdots - \beta_p B^p$$

represents the characteristic polynomial of the AR part and

$$\psi(B) = 1 + \alpha_1 B + \cdots + \alpha_q B^q$$

is the characteristic polynomial of the MA part. $\{\epsilon_t\}$ is a white noise with zero mean and a finite variance σ^2. We assume that $\phi(B)$ and $\psi(B)$ have no common factors and all roots of $\phi(B) = 0$ and $\psi(B) = 0$ are outside the unit circle, so the ARMA (p, q) is causal and invertible.

The autocovariances function (acf) of an ARMA (p, q) process is important and given by

$$V = \sum_{k=-\infty}^{\infty} \gamma(k) = \left(\frac{1 + \sum_{j=1}^{q} \alpha_j}{1 + \sum_{i=1}^{p} \beta_i}\right)^2 \sigma^2 \tag{3.4}$$

Under the assumption of stationary and invertible, as $k \to \infty$, the autocovariances are always absolutely summable and infinite, i.e. $\sum_{k=-\infty}^{\infty} |\gamma_X(k)| < \infty$. For simplicity, we use V to denote the sum of autovariances. The detailed discussion can be found in Hart, J.D. (1991).

Next, an overview of the development of nonparametric techniques will be described briefly. There are a large number of literatures on nonparametric regression analysis. According to the view of Li, Q. (2007), nonparametric methods were developed to be used in cases when the researcher knows nothing about a functional form for an object being estimated. Compared with the most widely used method, kernel regression, Local polynomial regression has a long history in the smoothing of time series, which can be traced back to 1931 by Macaulay. In the context of nonparametric regression the approach was introduced originally by Stone (1977), Cleveland (1979) and Katkovnik (1979) and investigated by many researchers. Lejeune (1985) and Müller (1987) studied certain equivalences between local polynomial estimators and

other kernel estimators, since local polynomial regression is a kind of kernel regression. Later, the work of Fan (1992) applied that local polynomial fitting has significant advantages over the Nadaraya-Watson regression estimator (Nadaraya, 1964 and Watson, 1964) and Gasser-Müller regression estimator (Gasser and Müller, 1979), since the local linear fitting in the univariate case can reduce the bias. Hastie and Loader (1993) showed that, for finite samples locally weighted regression is superior to kernel regression because of its intuitional definition and model unbiasedness. Significant modern contributions to this method include Silverman (1984), Härdle and Luckhaus (1984), Cleveland and Devlin (1988), Bönte and Fraiman (1988), Hastie and Tibshirani (1990), Fan (1992a, 1993), Fan and Gijbels (1992, 1995), Hastie and Loader (1993), Ruppert and Wand (1994), Fan, Gasser, Gijbels, Brockmann and Engel (1995), Härdle and Marron (1995), Hu and Huang (1996).

The remained part of this section is structured as follows; sub-section 3.2 gives a detailed introduction of kernel regression method for univariate data, including the relevant finite sample properties and asymptotic properties. Then a more attractive method- local polynomial regression that improved kernel regression will be discussed in the sub-section 3.3. This part will deal with local polynomial fitting, while locally weighted regression applied to decompose seasonal time series is the task of the next section. In addition, we will discuss the estimation and prediction of ARMA model.

3.2. Kernel Regression

3.2.1. Kernel Estimators

Kernel regression is the most widely used nonparametric approach. In 1964, Nadaraya and Watson proposed the Nadaraya-Watson kernel regression estimator independently. Gasser and Müller (1979) and Gasser, Jennen-Steinmetz and Engel (1991) introduced the alternative Gasser-Müller kernel regression estimator and showed that it is superior to Nadaraya-Watson

kernel regression estimator. Mack and Müller (1989) investigated the properties of these kernel estimators in the case of random design. Silverman (1986) talked about a useful introduction to kernel density estimation. Chu and Marron's proposal in 1991 given a detailed comparison and discussion of the properties of the most commonly used Nadaraya-Watson estimators and Gasser-Müller estimators. Other related discussions to kernel regression can be found in Härdle (1990, 1991), Fan (1992), Scott (1992), Wand and Jones (1995).

The focus of this paper on data-driven seasonal time series decomposition is based on locally weighted regression. As we known, it is a kind type of kernel regression and the locally weighted regression estimators are asymptotically to some kernel regression estimators. Lejeune (1985) and Müller (1987) studied certain equivalences between local polynomial estimators and other kernel estimators and it is possible to adapt some results of kernel regression to the locally weighted regression. Müller (1987) and Ruppert and Wand (1994) proposed that local polynomial estimators have the same asymptotic properties as kernel estimators. Hence, in order to better understand locally weighted regression, we give the main results of kernel regression in this part.

The kernel regression is just a direct extension and a weighted version of moving average, which is defined by

$$\hat{m}(x) = \sum_{t=1}^{n} w_t(x)\, Y_t \text{ with weights } w_t(x), \quad t = 1,2, \dots, n \qquad (3.5)$$

In order to describe the shape of the weights function by a density function, we use a kernel function K (u), which integrated to one. It is usually a symmetric density function and defined to be zero outside a certain range of u (usually on [-1, 1]).

$$\int K(u)du = 1$$

The weighting system by using kernel smoothers is defined by

$$w_t = \frac{K_h[x - x_t]}{\widehat{f_h}(x)} \tag{3.6}$$

Where

$$\widehat{f_h}(x) = n^{-1} \sum_{t=1}^{n} K_h(x - x_t)$$

denotes the Rosenblatt-Parzen kernel density estimator (Rosenblatt (1956); Parzen (1962)) of the marginal density of X and where

$$K_h(u) = h^{-1}K(u/h)$$

is a kernel function with a positive number h, usually called the bandwidth controlling the size of the local neighborhood. Too large h results in an over smoothed (or underfitted) estimate and causes a large modeling bias, while a too small h makes an undersmoothed (or overfitted) estimate and produce a large variance. Now we look at these bandwidth selection problem intuitively under two extreme scenarios, that is, what happens as $h \to 0$, and $h \to \infty$. Consider first $h \to 0$,

$$\widehat{m}(x_t) = \frac{K(0)Y_t}{K(0)} = Y_{t,} \qquad \widehat{m}(x) = 0, \quad x \neq x_t, \qquad t = 1, \cdots, n$$

shows that too small bandwidth reproduce the data, and other observations are estimated to be zero except $\widehat{m}(x_t)$. Then, in the case of $h \to \infty$, $K(\frac{x - x_t}{h}) \to K(0)$, and thus

$$\widehat{m}(x_t) \to \frac{n^{-1} \sum K(0)\, Y_t}{n^{-1} \sum K(0)} = n^{-1} \sum_{t=1}^{n} Y_t$$

Too large bandwidth leads to an oversmooth curve, which is composed of the average of observations. So an important essence of kernel smoothing is to find a suitable h, so that the squared bias and the variance of the estimates are in balance and the mean squared error (MSE) of the estimates is minimized. This problem will be carried out in section 5.

According to the view of Mays in 1995, a closer examination of the numerator of equation (3.6) lends some insight into the weighting scenario, that is, the points at locations close to x_t carry more information (more weight) about the value of m(x), in contrast, the points farther has less weight.

The form 3.6 has been proposed in 1964 by Nadaraya and Watson independently. The Nadaraya –Watson (NW) estimator is given by

$$\hat{m}_{NW}(x) = \frac{\sum_{t=1}^{n} Y_t K[(x - x_t)/h]}{\sum_{t=1}^{n} K[(x - x_t)/h]} = \sum_{t=1}^{n} w_t Y_t \qquad (3.7)$$

The weights are defined by

$$w_t = \frac{K[\frac{x - x_t}{h}]}{\sum_{t=1}^{n} K[\frac{x - x_t}{h}]}$$

and sum to one. For $|x - x_t| \leq h, w_t \geq 0$; $|x - x_t| > h, w_t = 0$.

In the next we look at another estimator proposed by Gasser and Müller in 1979. Hastie and Loader (1993) presented that the NW kernel estimator chooses weights by direct kernel evaluation, and the Gasser-Müller estimator does so by the convolution of the kernel. From the point of Fan, J. and Gijbels, I. (1996), when derivatives of the estimator and its asymptotic properties are taken, it is easy to find that the random denominator in (3.7) is inconvenient. So there is an assumption that the data have already been sorted as regards to x-variable. The Gasser-Müller (GM) estimator is defined by

$$\hat{m}_{GM}(x) = \frac{1}{h} \sum_{t=1}^{n} \int_{s_{t-1}}^{s_t} K\left[\frac{x - u}{h}\right] du \, Y_t \qquad (3.8)$$

Where s_t are mean points of the design. $s_0 = 0$, $s_t = (x_t + x_{t+1})/2$ for t=1, …, n-1 and $s_n = 1$ (Herrmann and Gasser, 1994). Note that there exists no denominator in the equation (3.8) because that the sum of the weights is one.

Until now, two different weighting schemes for kernel estimation have been introduced, referring to the NW kernel estimator and to the GM estimator. All kernel estimators have their relative merits. A public argument has risen about the most appropriate for use should be. Chu and Marron (1991) and Jones, Davis and Park (1994) have provided the comparative studies of the Nadaraya-Watson and Gasser-Müller estimators, including the presentation of their merit and demerit.

3.2.2. Kernel Functions

This part is mainly based on section 2.2 of Feng (1999). Besides bandwidth, the order of kernel regression function K plays an important role. Generally, k is assumed to be even, and the well known kernels are symmetric density functions, that is, kernel of order 2. A higher order of kernel will create an increase of the variance and the bias will go down. More results may be found in Gasser, Müller and Mammitzsch (1985) and Müller (1984). In this paper we require that K is symmetric with k=2.

In addition, most kernels are restricted to be compact supported due to following reasons:

1. Kernels with compact support have huge computational merit.
2. Simpler proofs with more powerful asymptotic results are allowed.
3. Compact support has been applied to optimal kernels and minimum variance kernels.
4. And over the whole interval, estimation using a kernel with non-compact support will be disturbed by boundary effects.

Without loss of generality it should be greater than -1 and smaller than 1, denote as supp (K) = [-1, 1].

Let k be a nonnegative even integer and $k \geq 2$. K is called a kernel of order k, $k \in \mathbb{N}$,if it is assumed to be Lipschitz continuous on the support(K) = $[-1, 1]$, and satisfies the following conditions

$$\int K(u)u^j du = \begin{cases} 1, & j = 0, \\ 0, & j = 1, \cdots, k-1, \\ (-1)^k (k!)\beta_k, & j = k, \end{cases}$$

where the kernel constant $\beta_k \neq 0$.

Commonly used second order kernel function:

$$K_\mu(u) = C_\mu (1-u^2)^\mu \mathbb{I}_{[-1,1]}(u) \qquad \mu = 0, 1, 2, 3, \cdots \tag{3.9}$$

Where $\mathbb{I}_{[-1,1]}(u)$ is the indicator function of the interval [-1, 1] and C_μ is defined by $\int K(u)du = 1$. According to the view of Müller (1984), " μ is called the degree of smoothness of a kernel function, which is an important characterization of a kernel function."

The kernels defined by (3.9) are the rectangular kernel for $\mu = 0$, the Epanechnikov kernel for $\mu = 1$, the Bisquare kernel for $\mu = 2$ and the triweight kernel for $\mu = 3$. Table 3.1 gives some commonly used second order kernels including above mentioned kernels and Gaussian kernel with $\mu = \infty$.

Note that the uniform kernel with $\mu = 0$ is the minimum-variance kernel owing to the minimal asymptotic variance. The Epanechnikov kernel with $\mu = 1$ is the optimal one since it has minimal asymptotic mean squared error (MSE).

Table 3.1: Selected second order kernels

Name	k	μ	Kernel
Uniform	2	0	$\frac{1}{2}\mathbb{I}_{[-1,1]}(u)$
Triangle	2	0	$(1-\lvert u\rvert)\mathbb{I}_{[-1,1]}(u)$
Epanechnikov	2	1	$\frac{3}{4}(1-u^2)\mathbb{I}_{[-1,1]}(u)$
Bisquare	2	2	$\frac{15}{16}(1-2u^2+u^4)\mathbb{I}_{[-1,1]}(u)$
Triweight	2	3	$\frac{35}{32}(1-3u^2+3u^4-u^6)\mathbb{I}_{[-1,1]}(u)$
Gaussian	2	∞	$\frac{1}{\sqrt{2\pi}}\exp(-\frac{1}{2}u^2)$

Source: Yuanhua, Feng, Kernel- and Locally Weighted Regression, Verlag für Wissenschaft und Forschung, 1999, p. 13

The estimation of derivatives of m(x) is also of great importance. In order to solve the bandwidth selection problem in section 5, for example, to apply the plug-in method we need to estimate $\int m^{(k)}(x)^2 dx$. The most popular estimator of $m^{(\nu)}(x)$ is proposed by Gasser and Müller in 1984, which is an extension of (3.8). A kernel function K_ν of order (ν, k) for estimating the νth derivative is defined as a Lipschitz continuous function satisfying the following condition:

$$\int K_\nu(x)x^j dx = \begin{cases} 0, & j = 0, \cdots, \nu-1, \nu+1, \cdots, k-1, \\ (-1)^\nu \nu!, & j = \nu, \\ (-1)^k (k!)\beta_{\nu,k}, & j = k, \end{cases}$$

where the kernel constant $\beta_{\nu,k} \neq 0$. Generally, it is assumed that $0 \leq \nu \leq k-2$, ν and k are both even or odd. If $k \geq \nu + 2$, K_ν is called a higher order kernel. A kernel estimator of the νth derivative of m(x) is defined by

$$\hat{m}^{(\nu)}(x) = \frac{1}{h^{\nu+1}} \sum_{t=1}^{n} \int_{s_{t-1}}^{s_t} K_\nu[(x-u)/h] du\, Y_t \qquad (3.10)$$

s_t are defined as (3.8). Note that (3.10) is the same as (3.8) in the case of $\nu = 0$.

3.2.3. Asymptotic Properties

Next we analyze some asymptotic properties of the bias and the variance of kernel regression as the sample number increase, and then we can derive the asymptotically local and global optimal bandwidth by minimizing the asymptotic mean squared error (MSE) and the asymptotic mean integrated squared error (MISE). To derive that given following assumptions required:

Assumption 1: m(x) is continuous on [0, 1] and at least p-times differentiable.
Assumption 2: The weight function K is a kernel of order (ν, k).Usually, k=2.
Assumption 3: The bandwidth satisfies $h \to 0, nh^{2\nu+1} \to \infty, \text{as } n \to \infty$.
Assumption 4: The design density f(x) is Lipschitz continuous on [0, 1].

The general global criterion for the goodness of fit is the weighted MISE defined by

$$\text{MISE} = \int_c^d f(x)\text{MSE}(x)dx = \int_c^d f(x)E[\hat{m}^{\nu}(x) - m^{(\nu)}(x)]^2 dx$$

Which is the integral of the error criterion MSE at an interval 0<c<d<1,

$$\text{MSE}=E[\hat{m}^{(\nu)}(x) - m^{(\nu)}(x)]^2$$

To construct the estimate of the MSE, it has been decomposed into the bias part and the variance part (Müller, 1988),

$$\text{var}(\hat{m}^{(\nu)}(x)) = \sigma^2 \sum_{t=1}^{n} w_t^2(x) = \frac{\sigma^2}{h^{2\nu+1} nf(x)} (R(K_\upsilon) + o(1))$$

$$B(\hat{m}^{\nu}(x)) = E[\hat{m}^{(\nu)}(x)] - m^{\nu}(x) = h^{k-\nu} m^{(k)}(x)\beta_{\nu,k} + O\left(\frac{1}{nh^{\nu}}\right) + o(h^{k-\nu})$$

where the kernel constant $\beta_{\nu,k} \neq 0$ and $R(K_\upsilon) = \int K_\upsilon(u)^2 du$.

So the asymptotic MSE can be written as

$$\mathrm{MSE}(x) = \mathrm{var}(\hat{m}^{(\nu)}(x)) + B(\hat{m}^{(\nu)}(x))^2$$

$$= \frac{\sigma^2}{nh^{2\nu+1}f(x)}\left(R(K_\upsilon(u)) + o(1)\right) + h^{2(k-\nu)}\left(\left[m^{(k)}(x)\beta_{\nu,k}\right]^2 + o(1)\right)$$

So we can obtain the asymptotically local optimal bandwidth by minimizing the MSE.

$$h_L(x) = \left(\frac{2\nu+1}{2(k-\nu)}\frac{\sigma^2 f(x)^{-1}}{\{m^{(k)}(x)\}^2}\frac{R(K_\upsilon)}{\beta_{\nu,k}^2}\frac{1}{n}\right)^{1/(2k+1)}$$

Furthermore, the MISE is defined by

$$\mathrm{MISE} = \frac{\sigma^2}{nh^{2\nu+1}}R(K_\upsilon)[(d-c) + o(1)] + h^{2(k-\nu)}\int_c^d f(x)\left[m^{(k)}(x)\right]^2 dx\left(\beta_{\nu,k}^2 + o(1)\right)$$

Similarly, the asymptotically global optimal bandwidth is

$$h_A = \left(\frac{2\nu+1}{2(k-\nu)}\frac{\sigma^2(d-c)}{\int_c^d f(x)\,[m^{(k)}(x)]^2 dx}\frac{R(K_\upsilon)}{\beta_{\nu,k}^2}\frac{1}{n}\right)^{1/(2k+1)}$$

Generally we use the kernel of order 2 and assume the design density is equidistant for fixed design, that is, $f(x) \equiv 1$ at $0<x<1$ or $0<c<d<1$ to avoid the boundary effect, $\nu = 0, k = 2$. So we can obtain the following results:

(1) The asymptotic bias: $B(m^{(\nu)}(x)) \approx h^2\beta_2\left(\frac{m''(x)}{2} + \frac{m'(x)f'(x)}{f(x)}\right)$, $\beta_2 = \int_{-1}^{1} u^2K(u)du$

(2) The asymptotic variance: $\mathrm{var}(m^{(\nu)}(x)) \approx \frac{\sigma^2}{hnf(x)}R(K_2)$, $R(K_2) = \int_{-1}^{1} K^2(u)du$

(3) The asymptotic MSE:

$$\mathrm{MSE}(x) = \mathrm{var}\left(m^{(\nu)}(x)\right) + B^2\left(m^{(\nu)}(x)\right) \approx \beta_2^4h^4 + \frac{[m''(x)]^2}{4} + \frac{\sigma^2}{hn}R(K_2)$$

(4) The asymptotic MISE:

$$\mathrm{MISE} = \int_c^d \mathrm{MSE}(x)dx \approx \beta_2^4 h^4 \frac{\int_c^d [m''(x)]^2 dx}{4} + \frac{\sigma^2}{hn} R(K_2)[d-c]$$

(5) The asymptotically local optimal bandwidth:

$$h_{opt} \approx \left(\frac{\sigma^2 R(K_2)}{\beta_2^4 [m''(x)]^2} \right)^{1/5} n^{-1/5} = C_{opt}(x) n^{-1/5} = h_L(x)$$

(6) The asymptotically global optimal bandwidth:

$$h_{opt} \approx \left(\frac{\sigma^2 R(K_2)[d-c]}{\beta_2^4 \int_c^d [m''(x)]^2 dx} \right)^{1/5} n^{-1/5} = C_{opt}(x) n^{-1/5} = h_A$$

Remark 1: The asymptotic bias depends on h, that is, the larger h the larger asymptotic bias. It is of the order $O(h^2)$. Also it depends on f(x) and $m''(x)$. Note that the asymptotic bias at the boundary point is of a larger order h, not h^2.

Remark 2: The asymptotic variance is of the order of $O(nh)^{-1}$.The larger h the smaller var(x). Besides, f(x) has compact on it, the larger f(x) the smaller variance.

Remark 3: The asymptotically local and global optimal bandwidths depend on unknown quantities σ^2 and $m^k(x)$. The larger σ^2 means that more observations have been used, so that the optimal bandwidths will be larger. Conversely, the larger $\left| m^{(k)}(x) \right|$ the fewer observations used, so the optimal bandwidth will be smaller. $C_{opt}(x)$ is often may be unknown and very complex.

The above applied kernel estimator is the NW-estimator; in the following we will give a short comparative of the NW-estimator and the GM-estimator. First, according to the analysis of Gasser, Jennen-Steinmetz and Engel (1991), in the special case $\nu = 0, k = 2$ the asymptotic bias of the GM-estimator is $h^2 \beta_2 \frac{m''(x)}{2}$ and the asymptotic variance of the GM-estimator is $\frac{C\sigma^2}{hnf(x)} R(K_2)$, where the factor C is 1 for fixed design and 1.5 for random design. Compared

with the asymptotic results of NW-estimator, by using the plug-in bandwidth selection method, the estimation of GM-estimator is easier and much more practical than that of NW-estimator, because that the second term of the NW-estimator is harder to calculate. Further, though none of the estimators is consistently optimal, but for regular fixed designs the GM-estimator is superior to the NW-estimator (Gasser, Jennen-Steinmetz and Engel, 1991). Indeed, both of them suffer from boundary effects. To solve this, we use a better approach-local linear estimator, which will be introduced in the next section; its bias is always of order h^2.

In addition, the above results are applied to the i.i.d case; next we consider the properties in the case of ARMA error. As we known, V stands for autocovariance, that is, $V = \sum_{k=-\infty}^{\infty}|\gamma_X(k)|$ (see Hart, J.D.1991). And it has influence only on the asymptotic variance, so σ^2 can be replaced by V, that is, $\mathrm{var}(\hat{m}^{(\nu)}(x)) = \frac{\mathbf{V}}{h^{2\nu+1}nf(x)}(R(K_\upsilon) + o(1))$.

Similarly, we can obtain the asymptotic MSE, the asymptotic MISE and the asymptotically optimal bandwidth by replacing σ^2 with V,

- $\mathrm{MSE} = \frac{\mathbf{V}}{nh^{2\nu+1}f(x)}\left(R(K_\upsilon(u)) + o(1)\right) + h^{2(k-\nu)}\left(\left[m^{(k)}(x)\beta_{\nu,k}\right]^2 + o(1)\right)$
- $h_L = \left(\frac{2\nu+1}{2(k-\nu)}\frac{\mathbf{V}f(x)^{-1}}{\{m^{(k)}(x)\}^2}\frac{R(K_\upsilon)}{\beta_{\nu,k}^2}\frac{1}{n}\right)^{1/(2k+1)}$
- $\mathrm{MISE} = \frac{\mathbf{V}}{nh^{2\nu+1}}R(K_\upsilon)[(d-c) + o(1)] + h^{2(k-\nu)}\int_c^d f(x)\left[m^{(k)}(x)\right]^2 dx\left(\beta_{\nu,k}^2 + o(1)\right)$
- $h_A = \left(\frac{2\nu+1}{2(k-\nu)}\frac{\mathbf{V}(d-c)}{\int_c^d f(x)\left[m^{(k)}(x)\right]^2 dx}\frac{R(K_\upsilon)}{\beta_{\nu,k}^2}\frac{1}{n}\right)^{1/(2k+1)}$

3.3. Local Polynomial Regression

3.3.1. Advantages of Local Polynomial Fitting

Compared with the most widely used nonparametric method-kernel regression, local regression has a long tradition and became the most attractive and applicable nonparametric method both from theoretical and practical point of view. The commonly used kernel estimators-NW estimator and GM estimator suffer from some drawbacks; first, both of them have a large order of bias when estimating at a boundary area. Second, the NW-estimator has zero minimax efficiency. Third, the NW-estimator leads to a large bias, while the GM-estimator can correct the problem, but the GM-estimator has to pay a price in variance in the case of random design.

The local polynomial fitting has some advantages as follows, which can overcome the drawbacks of kernel estimators. This method applies to all types of designs such as random and fixed design. Moreover, it addresses the boundary problem, the bias at the boundary have the same order as in the interior. Further, no boundary modifications are required if the ν-th derivative is to be estimated with $p - \nu$ odd, and the bias is of order $O(h^{p+1})$ at the whole interval. Indeed, it has nice minimax efficiency properties. Also, the interpretation and implementation are easy to understand and for derivative estimation this method has certain advantage over kernel regression. Hence, this method is called “golden-rule” of nonparametric technique. In the following the features of this method will be explained in detail.

3.3.2. Local Polynomial Estimators

Local polynomial regression, which based on Taylor expansion, improved kernel regression. Local polynomial estimators estimate the regression function at a particular point by “locally” fitting a pth degree polynomial to the data via weighed least squares. The Nadaraya-Watson

estimator is just a special form of local polynomial regression as local constant fit, it means that the order of polynomial is equal to zero. Of particular importance and simplicity is the local linear estimator, corresponding to p=1.

Consider the equidistant design regression model (3.1). Suppose that the unknown regression function m(x) is at least (p+1)-times differentiable at a given point x_0 , we can then approximate m(x) locally around x_0 by a polynomial of order p by using Taylor's expansion in a neighborhood of x_0 :

$$m(x) = m(x_0) + m'(x_0)(x - x_0) + \frac{1}{2}m''(x_0)(x - x_0)^2$$

$$+ \cdots + \frac{1}{p!}m^p(x_0)(x - x_0)^p + R_{p+1} \qquad (3.11)$$

Where R_{p+1} is a rest term of order $O[(x - x_0)^{p+1}]$.

This leads to the weighted least squares regression problem: minimize

$$\sum_{t=1}^{n}\left\{Y_t - \sum_{j=0}^{p}\beta_j(x_t - x_0)^j\right\}^2 K\left(\frac{x_t - x_0}{h}\right) \qquad (3.12)$$

Where h is bandwidth, K is a symmetric probability density function (a kernel of order two) with compact support [-1, 1]. The solution of minimize (3.12) is the estimated coefficient for local polynomial regression fit at x_0 , denote as $\widehat{\beta_j}$, j=0… p. The Taylor's exposition suggests that an estimator of $m^{(\nu)}(x)$ is

$$\hat{m}^{(\nu)}(x_0) = \nu!\,\widehat{\beta_\nu} \qquad (3.13)$$

If we work with matrix, it will be more convenient and intuitive.

Let $\mathbf{X} = \begin{bmatrix} 1 & x_1 - x_0 & \cdots & (x_1 - x_0)^p \\ \vdots & & \ddots & \vdots \\ 1 & x_n - x_0 & \cdots & (x_n - x_0)^p \end{bmatrix}$, which is the $n \times (p + 1)$ matrix of regressors, and put

$$y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \quad , \quad \beta(x_0) = \begin{pmatrix} \beta_0(x_0) \\ \vdots \\ \beta_p(x_0) \end{pmatrix}$$

Further, let K is the n × n diagonal matrix of weights:

$$\mathbf{K}(x_0) = \mathrm{diag}(k_t(x_0))$$

with $k_t(x_0) = K\left(\frac{x_t - x_0}{h}\right)$ as its ith diagonal entry.

Following the procedures of weighted least squares, (3.11) can be written as

$$\sum_{t=1}^{n} (Y_t - \mathbf{x}^T(x_t)\beta(x_0))^2 \, K\left(\frac{x_t - x_0}{h}\right) \Longrightarrow \min$$

the solution is given by

$$\widehat{\beta}(x_0) = (\mathbf{X}^T\mathbf{K}(x_0)\mathbf{X})^{-1}\mathbf{X}^T\mathbf{K}(x_0)\mathbf{y}$$

Hence we can obtain the local polynomial fitting at x_0 is

$$\begin{aligned} \widehat{m}(x_0) &= \mathbf{x}^T(x_0)\,\widehat{\beta}(x_0) \\ &= \mathbf{x}^T(x_0)(\mathbf{X}^T\mathbf{K}(x_0)\mathbf{X})^{-1}\mathbf{X}^T\mathbf{K}(x_0)\mathbf{y} \\ &= \mathbf{w}^T(x_0)\mathbf{y} \end{aligned}$$

where $\mathbf{w}^T(x_0) = (\mathbf{X}^T\mathbf{K}(x_0)\mathbf{X})^{-1}\mathbf{X}^T\mathbf{K}(x_0)$ is the local weights.

The νth derivative of m(x) can be easily written as

$$\widehat{m}^{(\nu)}(x_0) = \{\mathbf{w}^{\nu}(x_0)\}^T\mathbf{y}$$

with the weighting system $\{\mathbf{w}^{\nu}(x_0)\}^T$.

Furthermore, let $\mathbf{e}_{\nu+1} = (0, \cdots, 0, 1, 0, \cdots, 0)$ is the $(p+1) \times 1$ unit vector with a 1 in its $(\nu+1)$th entry and 0's elsewhere, the local least squares estimator $\widehat{m}^{(\nu)}(x)$ can be written as

$$\hat{m}^{(\nu)}(x) = \nu!\, \mathbf{e}_{\nu+1}{}^{T}(\mathbf{X}_x^T \mathbf{K}(x)\mathbf{X}_x)^{-1}\mathbf{X}_x^T \mathbf{K}(x)\mathbf{y} = \{\mathbf{w}^{\nu}(x)\}^T \mathbf{y} \tag{3.14}$$

with $\mathbf{X}_x = \begin{bmatrix} 1 & x_1 - x & \cdots & (x_1 - x)^p \\ \vdots & & \ddots & \vdots \\ 1 & x_n - x & \cdots & (x_n - x)^p \end{bmatrix}$.

3.3.3. Finite Sample Properties

A local polynomial estimator $\hat{m}^{(\nu)}(x)$ is a linear smoother with the weighting system $\mathbf{w}^{\nu}(x)$. Next, in order to investigate the finite sample properties, we introduce an equivalent definition which is defined by Müller (1987).

$$\hat{m}^{(\nu)}(x) = \{\mathbf{w}^{\nu}(x)\}^T \mathbf{y} = \sum_{t=1}^{n} w_t{}^{\nu}(x)\, Y_t$$

with the weighting system $\mathbf{w}^{\nu}(x) = (w_1^{\nu}(x), \dots, w_n^{\nu}(x))^T, x \in [0,1]$.

These weights $w_t^{\nu}(x), t = 1, \dots, n$, are determined by the following requirements:

1. Only the observations falling within the window [x- h, x+ h] will be used and have non-zero weights, so the estimator is localized.
2. For some given $k \geq 1$, the estimator is unbiased, if the unknown function is a polynomial of degree less than k (respectively, the corresponding derivative of order $\nu \leq k - 1$).
3. Minimized weighted sum of squares of the non-zero weights.

It will be intuitive, if we describe these conditions by using the formulas:

1. $w_t^{\nu}(x) = 0, \qquad \text{if } |x_t - x| > h,$
2. $\sum_{t=1}^{n} w_t{}^{\nu}(x)\,(x_t - x)^j = \begin{cases} 0 & 0 \leq j < k,\ j \neq \nu, \\ \nu! & j = \nu\ , \end{cases}$
3. $\tilde{\mathbf{w}}^T \tilde{\mathbf{K}}^{-1} \tilde{\mathbf{w}} = \min!$ with respect to $\tilde{w}$,

where $\tilde{\mathbf{w}}$ and $\tilde{\mathbf{K}}$ are the non-zero parts of $\mathbf{w}^{\nu}$ and $\mathbf{K}$.

It is obvious that property 1 only holds for compactly supported weight functions, but properties 2 and 3, besides for that has mentioned above, are also fit for non-compactly supported weight functions. Property 2 is called the model unbiasness of local weighted regression. Property 3 says that the local polynomial estimator is optimal following the weighted least squares idea. Properties 2 and 3 provide the evidence to certify the main difference between a local polynomial estimator and a kernel estimator. For a kernel estimator, these properties only hold approximately.

3.3.4. Asymptotic Properties

It is well known that a local polynomial regression is just a type of kernel regression. According to the study of asymptotic equivalence between the locally weighted regression and kernel regression by Lejeune (1985) and Müller (1987), Fan (1992) and Ruppert and Wand (1994), a local polynomial estimator is asymptotically equivalent to a certain kernel estimator, called an asymptotically equivalent kernel estimator of the local polynomial estimator. Hence, we can adapt asymptotic properties of kernel regression to locally weighted regression.

Recall that polynomial fits with $p - \nu$ odd is more preferable, in this case, the local polynomial estimator has automatic boundary correction, so that the kernel is overall of the order $k = p + 1$. However, if $p - \nu$ is even, the order in the interior is $k = p + 2$ with boundary problem. Most generally, $p - \nu$ is taken as 1.

Indeed, we assume that the asymptotic properties of $\widehat{m}^{(\nu)}(x)$ will be given at any point $t = ch, c \in [0,1]$, so the truncated kernel is defined by

$$K_c(u) = (\int_{-c}^{1} K(x)dx)^{-1} K(u) \mathbb{I}_{[-c,1]}(u)$$

For any ν, k and c, the asymptotically equivalent kernel for estimating $\hat{m}^{(\nu)}(x)$ is denoting by $K_{\nu,k,c}(u)$,

$$K_{\nu,k,c}(u) = \nu! \sum_{j=1}^{k} a_{\nu+1,j,c}\, u^{(j-1)} K_c(u)$$

See Gasser and Müller (1979).

The formula should satisfy following conditions:

$$\int_{-c}^{1} u^j K_{\nu,k,c}(u)du = \begin{cases} 0, & j = 0, \dots, \nu-1, \nu+1, \dots, k-1, \\ \nu!, & j = \nu, \\ \beta, & j = k, \end{cases}$$

Put the weighting system $w_t{}^{\nu}(x) = (w_1{}^{\nu}, \dots, w_n{}^{\nu})^T$ of a kernel estimator with $K_{\nu,k,c}(u)$. In the case of equidistant design we can define a kernel estimator of $\hat{m}^{(\nu)}(x)$ at x_0

$$\hat{m}^{(\nu)}(x) = \sum_{t=1}^{n} w_t{}^{\nu}(x)\, Y_t$$

Where $w_t{}^{\nu}(x) = \frac{1}{nh^{1+\nu}} K_{\nu,k,c}\left(\frac{x_t - x_0}{h}\right)$

To derive the asymptotic properties, we need some assumptions similar to that of kernel regression given in sub-section 3.2.3.

Assumption 1: m(x) is continuous on [0, 1] and at least p-times differentiable with

$p \geq \nu + 1$ and $p - \nu$ odd.

Assumption 2: A local polynomial fit of order $p = k - 1$ is used.

Assumption 3: The weight function K is a symmetric density with compact support [-1, 1].

Assumption 4: The bandwidth satisfies $h \to 0, nh^{2\nu+1} \to \infty, \text{as } n \to \infty$.

Under these assumptions we can obtain following results. First, the asymptotic bias is given by

$$B\left(\hat{m}^{(\nu)}(x)\right) = E\left(\hat{m}^{(\nu)} - m^{(\nu)}\right) \approx h^{(k-\nu)} \frac{m^{(k)}(x)\beta}{k!} + o\left(h^{(k-\nu)}\right)$$

And the asymptotic variance is given by

$$var[\hat{m}^{(\nu)}(x)] \approx \frac{\sigma^2}{nh^{1+2\nu}} (R(K_2) + o(1))$$

Then, we have the asymptotic MSE as follows:

$$MSE(x) \approx \beta_2^2 h^{2(k-\nu)} \frac{[m^{(k)}(x)]^2}{(k!)^2} + \frac{\sigma^2}{nh^{1+2\nu}} R(K_2)$$

As the mean integrated squared error, the asymptotic MISE is given by

$$MISE = \int_c^d MSE(x)dx \approx \beta_2^2 h^{2(k-\nu)} \frac{\int_c^d [m^{(k)}(x)]^2 dx}{(k!)^2} + \frac{\sigma^2}{nh^{1+2\nu}} R(K_2)[d-c]$$

It can be computed over the whole support [0, 1].

Minimizing the MISE, we can obtain the asymptotically global optimal bandwidth (only for v=0):

$$h_{opt} \approx \left(\frac{(k!)^2}{2k} \frac{\sigma^2 R(K_2)[d-c]}{\beta_2^2 \int_c^d [m^{(k)}(x)]^2 dx} \right)^{-1/2k+1} n^{-1/2k+1} = C_{opt}(x) n^{-1/2k+1}, k = 2, 4$$

Remark 1: Local polynomial regression is design-adaptive and overcomes the boundary problem partly. It is easy to understand and compute.

Remark 2: The asymptotic bias is of the order $O(h^{(k-\nu)})$, in the common case of $p - \nu = 1$, it is always of the order $O(h^2)$. And the asymptotic variance is of the order $O(nh^{1+2\nu})$. Both of them are getting larger and of a lower order for $\nu > 0$.

Remark 3: The asymptotically global optimal bandwidth has the relationship with σ^2 and $m^{(k)}(x)$ such as that of kernel regression. In addition, it also depends on k, that is, a larger k cause that the order of h gets larger.

Remark 4: Similarly, we can also replace σ^2 with V in the case of ARMA error. The results are showed as follows:

- $\mathrm{var}[\hat{m}^{(\nu)}(x)] \approx \frac{\mathbf{V}}{nh^{1+2\nu}}(R(K_2) + o(1))$
- $\mathrm{MSE}(x) \approx \beta_2^2 h^{2(k-\nu)} \frac{[m^{(k)}(x)]^2}{(k!)^2} + \frac{\mathbf{V}}{nh^{1+2\nu}} R(K_2)$
- $\mathrm{MISE} \approx \beta_2^2 h^{2(k-\nu)} \frac{\int_c^d [m^{(k)}(x)]^2 dx}{(k!)^2} + \frac{\mathbf{V}}{nh^{1+2\nu}} R(K_2)[d-c]$
- $h_{\mathrm{opt}} \approx \left(\frac{(k!)^2}{2k} \frac{\mathbf{V} R(K_2)[d-c]}{\beta_2^2 \int_c^d [m^{(k)}(x)]^2 dx} \right)^{-1/2k+1} n^{-1/2k+1} = C_{\mathrm{opt}}(x) n^{-1/2k+1}, k = 2, 4$

3.4. Estimation of ARMA Model

In this section we will present the theoretical concept for estimating the unknown quantities of ARMA model, which is useful for selecting the optimal bandwidth in section 5. First we introduce the estimation of AR model, since AR model is easily to estimate, and then the estimation of ARMA model will be given.

3.4.1. Estimation of the AR Model

For simplicity we first estimate the AR (1) model

$$X_t = \beta X_{t-1} + \epsilon_t$$

Where $|\beta| < 1$ and ϵ_t is a white noise process. In this model we have two unknown parameters β and σ_ϵ^2. For estimating β, we have the relation $\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \beta^{|k|}$. Note that the autocovariance $\gamma(k)$ is calculated by

$$\gamma(k) = \frac{\beta^{|k|}}{1-\beta^2}\sigma_\epsilon^2$$

and the estimator of $\gamma(k)$ is

$$\hat{\gamma}(k) = \frac{1}{n}\sum_{t=1}^{n-k}(X_t - \overline{X})(X_{t+k} - \overline{X})$$

In the case of AR (1), we have simply $\beta = \rho(1)$. Hence an estimator of $\rho(1)$ is also the estimator of β.

$$\hat{\beta} = \hat{\rho}(1) = \frac{\hat{\gamma}(1)}{\hat{\gamma}(0)} = \frac{\sum_{t=1}^{n-1}(x_t - \bar{x})(x_{t+1} - \bar{x})}{\sum_{t=1}^{n}(x_t - \bar{x})^2}$$

If the mean of observations is zero, we have

$$\hat{\beta} = \hat{\rho}(1) = \frac{\sum_{t=1}^{n-1} x_t x_{t+1}}{\sum_{t=1}^{n} {x_t}^2}$$

Then we consider the estimation of σ_ϵ^2. Form the formula of autocovariance as mentioned above we can obtain

$$\gamma(0) = \frac{1}{1-\beta^2}\sigma_\epsilon^2$$

By definition, the estimator of σ_ϵ^2 is

$$\hat{\sigma}_\epsilon^2 = \hat{\gamma}(0)(1-\hat{\beta}^2)$$

with $\hat{\gamma}(0) = \frac{1}{n}\sum_{t=1}^{n} x_t^2$.

For an AR (p) model with unknown mean μ

$$X_t - \mu = \beta_1(X_{t-1} - \mu) + \cdots + \beta_p(X_{t-p} - \mu) + \epsilon_t$$

Then we define $Y_t = X_t - \mu$, $X_{1t} = X_{t-1} - \mu, \ldots, X_{pt} = X_{t-p} - \mu$, hence we get a linear regression

$$Y_t = \beta_1 X_{1t} + \cdots + \beta_p X_{pt} + \epsilon_t$$

The estimators of $\beta_1, \ldots, \beta_p$ can be obtained by the least squares methods.

First, the unknown mean μ is estimated by

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{t=1}^{n} x_t$$

And let $y_t = x_t - \bar{x}$ be the centralized data.

So the least squares estimators is obtained by minimizing

$$\sum_{t=p+1}^{n} (\hat{\epsilon}_t)^2 = \sum_{t=p+1}^{n} [y_t - \sum_{i=1}^{p} \hat{\beta}\, y_{t-i}]^2$$

with the approximation solutions (Yule-Walker Estimators)

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_p \end{pmatrix} = \begin{pmatrix} \hat{\gamma}(0) & \hat{\gamma}(1) & \ldots & \hat{\gamma}(p-1) \\ \hat{\gamma}(1) & \hat{\gamma}(0) & \ldots & \hat{\gamma}(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\gamma}(p-1) & \hat{\gamma}(p-2) & \ldots & \hat{\gamma}(0) \end{pmatrix}^{-1} \begin{pmatrix} \hat{\gamma}(1) \\ \hat{\gamma}(2) \\ \vdots \\ \hat{\gamma}(p) \end{pmatrix}$$

Also, we can replace $\hat{\gamma}(k)$ by $\hat{\rho}(k)$

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_p \end{pmatrix} = \begin{pmatrix} 1 & \hat{\rho}(1) & \dots & \hat{\rho}(p-1) \\ \hat{\rho}(1) & 1 & \dots & \hat{\rho}(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}(p-1) & \hat{\rho}(p-2) & \dots & 1 \end{pmatrix}^{-1} \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \\ \vdots \\ \hat{\rho}(p) \end{pmatrix}$$

To fit the AR (p) model, the values of the first p autocorrelations are required.

For example, for AR (2) model, we need to calculate the first two autocorrelations

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} 1 & \hat{\rho}(1) \\ \hat{\rho}(1) & 1 \end{pmatrix}^{-1} \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \end{pmatrix}$$

leads to the solution

$$\hat{\beta}_1 = \hat{\rho}(1)\left(\frac{1-\hat{\rho}(2)}{1-\hat{\rho}^2(1)}\right)$$

$$\hat{\beta}_2 = 1 - \frac{1-\hat{\rho}(2)}{1-\hat{\rho}^2(1)}$$

3.4.2. Estimation of the ARMA Model

For an ARMA (p, q) model with zero mean ($\hat{\mu} = 0$)

$$X_t = \beta_1 X_{t-1} + \cdots + \beta_p X_{t-p} + \alpha_1 \epsilon_{t-1} + \cdots + \alpha_q \epsilon_{t-q} + \epsilon_t$$
$$\Rightarrow \epsilon_t = X_t - \beta_1 X_{t-1} - \cdots - \beta_p X_{t-p} - \alpha_1 \epsilon_{t-1} - \cdots - \alpha_q \epsilon_{t-q}$$

We have unknown parameters $\sigma_\epsilon^2, \beta_1, \dots, \beta_p, \alpha_1, \dots, \alpha_q$.

For AR model, it is easy to calculate ϵ_t , whereas for MA and ARMA models, ϵ_t depends on unobservable $\epsilon_{t-1}, \dots, \epsilon_{t-q}$. Hence an approximation solution is given in the following:

1. Given time series $x_1, \dots, x_n$. Calculate $\hat{\mu} = \bar{x}$ and define $y_t = x_t - \bar{x}$.
2. Let $\hat{\epsilon}_t = 0, y_t = 0$ for $t \leq 0$.

3. Set $\hat{\epsilon}_1 = y_1$

$\hat{\epsilon}_2 = y_2 - \beta_1 y_1 - \alpha_1 \hat{\epsilon}_1$

$\vdots$

$\hat{\epsilon}_n = y_t - \beta_1 y_{t-1} - \cdots - \beta_p y_{t-p} - \alpha_1 \hat{\epsilon}_{t-1} - \cdots - \alpha_q \hat{\epsilon}_{t-q}$

for $t > \max(p, q)$.

Then we can estimate $\beta_1, \ldots, \beta_p, \alpha_1, \ldots, \alpha_q$ by minimizing the least squares criterion

$$Q = (\beta_1, \ldots, \beta_p; \alpha_1, \ldots, \alpha_q) = \sum_{t=\max(p,q)+1}^{n} \hat{\epsilon}_t^{\,2}$$

Furthermore, we introduce two well known information criterions for selecting parameters p and q. First we denote the minimized value of Q as Q (p, q), and then define

$$\hat{\sigma}_{p,q}^{\,2} = \frac{Q(p,q)}{n}$$

So $\hat{\sigma}_{p,q}^{\,2}$ is an estimator of σ_ϵ^2 by definition.

Then these two criterions are given as follows:

1. The AIC(Akaike's information criterion) proposed by Akaike in 1978:

   AIC (p, q) $:= \ln \hat{\sigma}_{p,q}^{\,2} + 2\frac{p+q}{n}$,

2. The BIC(Bayesian's information criterion) from Akaike 1977 and Schwarz 1978:

   BIC (p, q) $:= \ln \hat{\sigma}_{p,q}^{\,2} + \ln(n)\frac{p+q}{n}$.

Compared these two criterions, BIC is more preferable to AIC due to following reasons:

1. $\hat{p}(\text{AIC}) \geq \hat{p}(\text{BIC}),\ \hat{q}(\text{AIC}) \geq \hat{q}(\text{BIC})$, since $\ln(n)$ is larger than 2.
2. $\hat{p}(\text{AIC}) \nrightarrow p, \hat{q}(\text{AIC}) \nrightarrow q, \text{as } n \to \infty$. $\hat{p}, \hat{q}$ selected by AIC are not consistent.
3. $\hat{p}(\text{BIC}) \to p, \hat{q}(\text{BIC}) \to q, \text{as } n \to \infty$. $\hat{p}, \hat{q}$ selected by BIC are consistent estimators.

3.5. Forecasting for ARMA Model

After we fitted an AR model or ARMA model to the given time series $x_1, \cdots, x_n$, we can begin with the point or interval prediction for a further observation X_{n+k} with $k \geq 1$. Similar to the estimation of ARMA model, we first describe the point forecasting for AR model, since it is easier than MA or ARMA model. The method we will present is called Box-Jenkins Method proposed by Box Jenkins in 1970. Then the interval prediction process for an ARMA model will be given, if it is normally distributed. More detailed discussions of time series forecasts can be found in Hamilton (1994).

3.5.1. Point Forecasting for AR Model

Assume that we have an AR (p) model

$$X_n = \beta_1 X_{n-1} + \cdots + \beta_p X_{n-p} + \epsilon_n$$

We want to forecast X_{n+k}(i.e., k steps ahead from forecast origin n),

$$X_{n+k} = \beta_1 X_{n+k-1} + \cdots + \beta_p X_{n+k-p} + \epsilon_{n+k}$$

Then the MSE of the prediction, denoted by $\widehat{X}_{n+k}$ is

$$\mathrm{MSE}(\widehat{X}_{n+k}) = \mathrm{E}[(\widehat{X}_{n+k} - X_{n+k})^2]$$

The linear optimal prediction of X_{n+k} is defined by the conditional mean of X_{n+k}

$$\widehat{X}_{n+k} = \mathrm{E}[X_{n+k} | X_n, \cdots, X_1]$$

As a result, the minimum mean square error forecast for X_{n+k} can be presented by

$$\widehat{X}_{n+k} = \beta_1\left(\widehat{X}_{n+k-1}|X_n, \cdots, X_1\right) + \cdots + \beta_p(X_{n+k-p}|X_n, \cdots, X_1)$$

The prediction process is given as following:

1. One step ahead (k=1):

$$X_{n+1} = \beta_1 X_n + \cdots + \beta_p X_{n+1-p} + \epsilon_{n+1}$$

$$\widehat{X}_{n+1} = \beta_1 X_n + \cdots + \beta_p X_{n+1-p}$$

$$X_{n+1} - \widehat{X}_{n+1} = \epsilon_{n+1}$$

$$\text{var}(X_{n+1} - \widehat{X}_{n+1}) = \sigma_\epsilon^2$$

2. Two steps ahead (k=2):

$$X_{n+2} = \beta_1 X_{n+1} + \beta_2 X_n + \cdots + \beta_p X_{n+2-p} + \epsilon_{n+2}$$

$$\widehat{X}_{n+2} = \beta_1 \widehat{X}_{n+1} + \beta_2 X_n + \cdots + \beta_p X_{n+2-p}$$

$$X_{n+2} - \widehat{X}_{n+2} = \epsilon_{n+2} + \beta_1 \epsilon_{n+1}$$

$$\text{var}(X_{n+1} - \widehat{X}_{n+1}) = \sigma_\epsilon^2(1 + {\beta_1}^2)$$

In general, k steps ahead with k> p :

$$X_{n+k} = \beta_1 X_{n+k-1} + \cdots + \beta_p X_{n+k-p} + \epsilon_{n+k}$$

$$\widehat{X}_{n+k} = \beta_1 \widehat{X}_{n+k-1} + \cdots + \beta_p X_{n+k-p}$$

3.5.2. Interval Forecasting for ARMA Model

In the following we consider the interval prediction for a normal ARMA model, which can be represented as the function in infinite MA form.

$$X_t = \sum_{i=0}^{\infty} \alpha_i \epsilon_i$$

Then the conditional variance of X_{n+k} is

$$\text{var}(X_{n+k}|X_n, \cdots, X_1) = \sigma_\epsilon^2 \sum_{i=0}^{k-1} {\alpha_i}^2$$

where $\alpha_i = 1$.

An approximate 95% prediction interval for X_{n+k}:

$$\widehat{X}_{n+k} \pm 1{,}96\sigma_\epsilon \sqrt{\sum_{i=0}^{k-1} {\alpha_i}^2}$$

For one step ahead the interval is simply

$$\widehat{X}_{n+1} \pm 1{,}96\sigma_\epsilon$$

For two steps ahead the interval is

$$\widehat{X}_{n+2} \pm 1{,}96\sigma_\epsilon \sqrt{1 + {\alpha_1}^2}$$

For two steps ahead the interval is

$$\widehat{X}_{n+3} \pm 1{,}96\sigma_\epsilon \sqrt{1 + {\alpha_1}^2 + {\alpha_2}^2}$$

And for k steps ahead the interval is

$$\widehat{X}_{n+k} \pm 1{,}96\sigma_{\epsilon}\sqrt{1 + {\alpha_1}^2 + \cdots + {\alpha_{k-1}}^2}$$

Observe that the variance increase as k become larger.

Furthermore, $\lim \widehat{X}_{n+k} = E(X_{n+k})$, as $k \to \infty$. And from the forecast error

$$X_{n+k} - \widehat{X}_{n+k} = \epsilon_{n+k} + \alpha_1\epsilon_{n+k-1} + \cdots + \alpha_{k-1}\epsilon_{n+1}$$

We can see that because

$$\text{var}(X_{n+k}) = \sigma_{\epsilon}(1 + {\alpha_1}^2 + \cdots + {\alpha_{k-1}}^2),$$

$$\text{limvar}(X_{n+k}) = \sigma_{\epsilon}(1 + {\alpha_1}^2 + \cdots) = \text{var}(X_{n+k}).$$

The above results are applied to stationary time series; it will be more complicated for nonstationary time series, and we do not need to consider in this paper.

4. Semiparametric Regression Model with ARMA Errors and Seasonality

4.1. Time Series Decomposition with LWR

As so far we have presented the semiparametric regression model with ARMA errors without seasonality in detail. From now on, the seasonality will be considered. The aim of this section is to discuss the application of nonparametric techniques in seasonal time series. Since a long time smoothing and decomposing seasonal time series have been applied. In order to obtain a clearer structure of the economy, seasonal adjustment- the estimation and removal of the seasonal component become an important issue. This section is mainly based on the proposal of Heiler and Michels (1994) and Feng (1998).

The BV method is based on local least squares. Consider an equidistant time series Y_t, which is assumed to follow an additive components model

$$Y_t = m(x_t) + Z_t = G(x_t) + S(x_t) + Z_t, \qquad t = 1,2,\dots,n \tag{4.1}$$

where m=G+S is the mean function. $G(x_t)$ is the trend-cyclical component, $S(x_t)$ is the seasonal component with seasonal period s (i.e. s=12 for monthly series and s=4 for quarterly series), and Z_t is a stationary, invertible ARMA(p, q) error process with autocovariances $V = \sum_{k=-\infty}^{\infty} \gamma(k)$.

Following the BV method, assume that the trend- cyclical component $G(x_t)$ is at least (p+1) times continuously differentiable, so that it can be expressed in a Taylor series around a point t_0. Then we obtain a local polynomial model of order p

$$G(x_t) = \sum_{j=0}^{p} \beta_{1j}(t_0)(t - t_0)^j \tag{4.2}$$

Similarly, the seasonal component $S(x_t)$ can be estimated by a local finite Fourier expression

$$S(x_t) = \sum_{j=1}^{q} [\beta_{2j}(t_0)\cos\lambda_j(t - t_0) + \beta_{3j}(t_0)\sin\lambda_j(t - t_0)] \quad (4.3)$$

Where λ_j is the seasonal frequencies, $\lambda_1 = 2\pi/s$ and $\lambda_j = j\lambda_1$, for j= 2… q. besides, $q = [s/2]$ with [x] denotes the maximal integer which is not larger than x.

Let $\beta_1(t_0) = (\beta_{10}(t_0), \cdots, \beta_{1p}(t_0))^T$,

$\beta_2(t_0) = (\beta_{21}(t_0), \beta_{31}(t) \cdots, \beta_{2q}(t_0), [\beta_{3q}(t_0)])^T$,

$\beta(t_0) = (\beta_1(t_0)^T, \beta_2(t_0)^T)^T$,

$\mathbf{x}_1(t) = (1, (t - t_0), \ldots, (t - t_0)^p)^T$,

$\mathbf{x}_2(t) = (\cos\lambda_1(t - t_0), \sin\lambda_1(t - t_0), \ldots, \cos\lambda_q(t - t_0), [\sin\lambda_q(t - t_0)])^T$,

$\mathbf{x}(t) = (\mathbf{x}_1(t)^T, \mathbf{x}_2(t)^T)^T$,

where **X** is the $[n \times (p + s)]$-design matrix. The last terms of β_2 and $\mathbf{x}_2$, which are marked by [ ], are only applied to odd s, for even s they have to be omitted because of $\lambda_q = \pi$.

So $G(x_t)$ and $S(x_t)$ can be written as

$$G(x_t) = \mathbf{x}_1(t)^T\beta_1(t_0),$$
$$S(x_t) = \mathbf{x}_2(t)^T\beta_2(t_0),$$
$$m(x_t) = \mathbf{x}(t)^T\beta(t_0).$$

Next given the condition that the weighting function $K(u)$ is a second order kernel function with compact support [-1, 1] and h denotes the bandwidth. In addition, $\mathbf{y} = (Y_1, \cdots, Y_n)^T$ is the data vector and **K** denotes a $n \times n$ matrix

$$K = \begin{pmatrix} K_1 & 0 & \ldots & 0 \\ 0 & K_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & K_n \end{pmatrix}$$

with

$$k_t(x_0) = \begin{cases} K\left(\frac{t-t_0}{h}\right), & |t - t_0| \le h, \\ 0 & \text{otherwise}. \end{cases} \tag{4.4}$$

Furthermore, let $\mathbf{e}_j$ is the j-th $(p+1) \times 1$ unit vector, and $\mathbf{e}_1$ is just defined as $\mathbf{x}_1(t_0)$; $\mathbf{\Phi}_s = \mathbf{x}_2(t_0)$ denote a $(s-1) \times 1$ vector with 1 in its odd entries and 0 elsewhere.

Using the locally weighted squares criterion, $G(x_t), S(x_t)$ and $m(x_t)$ are estimated by minimizing

$$Q = \sum_{t=1}^{n} \{ Y_t - \sum_{j=0}^{p} \beta_{1j}(t - t_0)^j$$

$$- \sum_{j=1}^{q} \beta_{2j} cos\lambda_j(t - t_0) + [\beta_{3j} sin\lambda_j(t - t_0)]\}^2 K\left(\frac{t - t_0}{h}\right) \Rightarrow \min.$$

leads to the solutions

$$\hat{\beta}(t_0) = \begin{pmatrix} (\hat{\beta}_1(t_0) \\ (\hat{\beta}_2(t_0) \end{pmatrix}$$

$$= (\mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{X}(t_0))^{-1} \mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{y}$$

Then we have

$$\hat{m}(x_t) = \mathbf{x}(t)^T \hat{\beta}(t_0)$$

$$= \mathbf{x}(t)^T (\mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{X}(t_0))^{-1} \mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{y}$$

$$= \mathbf{x}_1(t)^T \hat{\beta}_1(t_0) + \mathbf{x}_2(t)^T \hat{\beta}_2(t_0)$$

$$= (\mathbf{x}_1(t)^T, \mathbf{0}^T)(\mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{X}(t_0))^{-1} \mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{y}$$

$$+ (\mathbf{0}^T, \mathbf{x}_2(t)^T) \left(\mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{X}(t_0)\right)^{-1} \mathbf{X}(t_0)^T \mathbf{K}(t_0) \mathbf{y}$$

$$= (\mathbf{e_1^T}, \mathbf{\Phi}_s^T)(\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{X}(t_0))^{-1}\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{y}$$

$$=: \mathbf{w}(t)^T\mathbf{y} \tag{4.5}$$

Similarly, we get

$$\widehat{G}(x_t) = (\mathbf{e}_1^T, \mathbf{0}^T)(\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{X}(t_0))^{-1}\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{y}$$

$$=: \mathbf{w}_1(t)^T\mathbf{y} \tag{4.6}$$

$$\widehat{S}(x_t) = (\mathbf{0}^T, \mathbf{\Phi}_s^T)\left(\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{X}(t_0)\right)^{-1}\mathbf{X}(t_0)^T\mathbf{K}(t_0)\mathbf{y}$$

$$=: \mathbf{w}_2(t)^T\mathbf{y} \tag{4.7}$$

where **0** is a vector of zeros of appropriate dimension.

Like local polynomial fitting, the ν-th derivative of G can be estimated with a local polynomial of order $p > \nu$ and $p - \nu$ odd. The common and simple case is $p - \nu = 1$, so the order of kernels generated is $k = p + 1$. Then we obtain the estimate of the ν-th derivative of G defined by

$$\widehat{G}^{(\nu)}(x_t) = (\nu!, \mathbf{e}_{1+\nu}^T, \mathbf{0}^T)(\mathbf{X}^T\mathbf{K}(t)\mathbf{X})^{-1}\mathbf{X}^T\mathbf{K}(t)$$

$$=: \mathbf{w}_1^{\nu}(t)^T\mathbf{y} \tag{4.8}$$

$\widehat{G}(x_t)$ is just a special case with $\nu = 0$.

The estimators $\widehat{m}$, $\widehat{G}$,$\widehat{S}$ and $\widehat{G}^{(\nu)}$ are all linear smoothers. The weighting systems of these estimators are

$$\mathbf{w} = (w_1, \dots, w_n)^T, \mathbf{w}_1 = (w_{11}, \dots, w_{1n})^T, \mathbf{w}_2 = (w_{21}, \dots, w_{2n})^T$$

with the relationships $\mathbf{w}(t) = \mathbf{w}_1(t) + \mathbf{w}_2(t)$. At any point t it holds that

$$\sum_{i=1}^{n} w_i(t) = 1\ ,\ \sum_{i=1}^{n} w_{1i}(t) = 1,$$

$$\sum_{i=1}^{n} w_{2i}(t) = 0 \text{ and } \sum_{i=1}^{n} w_i{}^{\nu}(t) = 0, \text{ for } \nu \geq 1.$$

4.2. Finite Sample Properties

In the following we will consider the finite sample properties. The estimators above have quite similar properties like the local polynomial fitting, which have been mentioned in sub-section 3.3.3. These properties also have restricted on the weighting systems, which are proposed by Müller (1987). We present these properties also by explanation and formula.

1. The weights will be zero, if we use the observations out of the window [x- h, x+ h].
2. The proposed estimators are bias free for a polynomial of degree less than k and for an exactly periodic function with the period s.
3. The weighted sum of squares of the non-zero weights is minimized.

The corresponding formulas are given below:

1. $w_i(t) = 0, \qquad \text{if } |i - t| > h,$
2. $\sum_{i=1}^{n} w_i(t)\,(i-t)/n)^j = \begin{cases} 1, & j = 0 \\ 0, & 1 \leq j < k \end{cases}$

2' $\begin{cases} \sum_{i=1}^{n} w_i(t)\cos(\lambda_j(i-t)) = 1 \\ \sum_{i=1}^{n} w_i(t)\sin(\lambda_j(i-t)) = 0 \end{cases} \quad j = 1,..,q$

3. $\tilde{\mathbf{w}}^T \tilde{\mathbf{K}}^{-1} \tilde{\mathbf{w}} = \min!$

$\tilde{\mathbf{w}}$ and $\tilde{\mathbf{K}}$ are defined as the same as that in section 3.3.3, they are the non-zero parts of $\mathbf{w}^{\nu}$ and $\mathbf{K}$.

4.3. Asymptotic Properties

In this section we will continue to investigate the asymptotic properties, which are useful for applying the plug-in bandwidth selection method.

Assumption 1: $G(x_t)$ is at least p+1 times differentiable.

Assumption 2: The bandwidth satisfies $h \to 0, nh \to \infty, \text{as } n \to \infty$.

Assumption 3: $S(x_t)$ is exactly periodic function with period s.

The first two conditions are similar to the nonparametric regression without seasonality in the last section. The third is a parametric assumption on the seasonal component, which is made for simplicity and can easily be relaxed to a general slowly changing seasonal component. Under the second assumption it can be showed that $\hat{G}$ is equivalent to some kernel estimator, in the following we denote this kernel estimator by $K_p(u)$.

To deal with $\hat{S}$, we will introduce a kernel estimator of S. let

$$Q_s(i) = \begin{cases} (s-1), & \text{if } \frac{i-t}{s} \text{ is an integer,} \\ -1 & \text{otherwise .} \end{cases} \tag{4.9}$$

And

$$\breve{w}_{2i} = (nh)^{-1} Q_s(i) K\left(\frac{x_i - x_t}{h}\right)$$

A kernel estimator of S is defined by

$$\breve{S}(x_t) = \sum_{i=1}^{n} \breve{w}_{2i}\, y_i =: \breve{\mathbf{w}}_2^T \mathbf{y} \tag{4.10}$$

$\{\breve{w}_{2i}\}$ are asymptotically periodic with the same period s. Suppose that corresponding boundary correction is done for $\breve{S}$, then it can be shown that, under the second assumption, $\breve{S}$ and $\hat{S}$ are asymptotically equivalent, too (Feng,1999).

As an error criterion for bandwidth selection we use the mean averaged squared error (MASE). Define $R(K) = \int_{-1}^{1} K^2(u)du$. Then we have following results.

(1) $\hat{S}(x_t)$ is unbiased and the biases of $\hat{m}(x_t)$ and $\hat{G}(x_t)$ are the same:

$$B[\hat{m}(x_t)] = B\left[\hat{G}(x_t)\right] = \frac{1}{(k!)}\left\{\left[\int u^k\, k_p(u)du\right]G^{(k)}(x_t)\right\}h^k$$

(2) The variances of $\hat{m}(x_t)$, $\hat{G}(x_t)$ and $\hat{S}(x_t)$ are given by

$$var[\hat{m}(x_t)] = (nh)^{-1}\sigma^2\{R(K_p) + (s-1)R(K)\}\{1 + O[(nh)^{-1}]\}$$

$$var\left[\hat{G}(x_t)\right] = (nh)^{-1}\sigma^2 R(K_p)\{1 + O[(nh)^{-1}]\}$$

$$var\left[\hat{S}(x_t)\right] = (nh)^{-1}\sigma^2(s-1)R(K)\{1 + O[(nh)^{-1}]\}$$

(3) The MASE of $\hat{m}(x_t)$ is

$$MASE\left(\hat{m}(x_t)\right) = \frac{1}{n}\sum_{t=1}^{n}[E(\hat{m}(x_t)) - m(x_t)]^2$$

$$= \frac{\sigma^2}{nh}\{R(K_p) + (s-1)R(K)\}$$

$$+ \frac{1}{(k!)^2}\left\{\int\left\{G^{(k)}(x)\right\}^2 dx\,[\int u^k K(u)du]^2\right\}h^{2k}$$

(4) Minimizing the MASE, we obtain the asymptotically optimal bandwidth

$$h_{opt} = \left(\frac{(k!)^2}{2k}\frac{\sigma^2\{R(K_p) + (s-1)R(K)\}}{\int\{G^{(k)}(x)\}^2 dx\left\{\int u^k\, K_p(u)du\right\}^2}\right)^{1/2k+1} n^{-1/2k+1}$$

Similarly, we can extend these results to the case of ARMA error process. We only need to replace σ^2 with V (see Hart, J.D.1991). So we have

(1) The bias is the same as i.i.d case.

$$B[\hat{m}(x_t)] = B\left[\hat{G}(x_t)\right] = \frac{1}{(k!)}\left\{\left[\int u^k\, k_p(u)du\right] G^{(k)}(x_t)\right\} h^k$$

(2) The variances of $\hat{m}(x_t)$, $\hat{G}(x_t)$ and $\hat{S}(x_t)$ are

$$var[\hat{m}(x_t)] = (nh)^{-1}\mathbf{V}\{R(K_p) + (s-1)R(K)\}\{1 + O[(nh)^{-1}]\}$$

$$var\left[\hat{G}(x_t)\right] = (nh)^{-1}\mathbf{V}R(K_p)\{1 + O[(nh)^{-1}]\}$$

$$var\left[\hat{S}(x_t)\right] = (nh)^{-1}\mathbf{V}(s-1)R(K)\{1 + O[(nh)^{-1}]\}$$

(3) And the MASE is

$$MASE\left(\hat{m}(x_t)\right) = \frac{1}{n}\sum_{t=1}^{n}[E(\hat{m}(x_t)) - m(x_t)]^2$$

$$= \frac{\mathbf{V}}{nh}\{R(K_p) + (s-1)R(K)\}$$

$$+ \frac{1}{(k!)^2}\left\{\int\left\{G^{(k)}(x)\right\}^2 dx\,[\int u^k K(u)du]^2\right\} h^{2k}$$

(4) The asymptotically optimal bandwidth

$$h_{opt} = \left(\frac{(k!)^2}{2k}\frac{\mathbf{V}\{R(K_p) + (s-1)R(K)\}}{\int\{G^{(k)}(x)\}^2 dx\left\{\int u^k\, K_p(u)du\right\}^2}\right)^{1/2k+1} n^{-1/2k+1}$$

The unknown quantities in the formula are the variance σ^2, the sum of autocovariance V, and the function $\int\left\{G^{(k)}(x)\right\}^2 dx$, which is denoted as I for simplicity. So the issue of bandwidth selection problem is the estimation of these unknown quantities, which will be presented in the next section.

5. Bandwidth Selection

5.1. Introduction

As we known, the choice of bandwidth is crucial in the nonparametric technique, since the magnitude of bandwidth controls the degree of smoothness. Choosing a too large bandwidth may lead to a large bias, whereas selecting a too small bandwidth may cause a large variance. In order to choose the asymptotical optimal bandwidth, a variety of methods have been developed.

In 1996, Jones, Marron and Sheather grouped these methods into “first generation” and “second generation”. Most first generation methods were developed before 1990 and contributions to the context of nonparametric density estimation can be found in the proposal of Rice (1984), Silvemann (1986), Härdle, Hall and Marron (1988) and Scott (1992). One widely studied method is the cross-validation method proposed by Clark (1975) and Wahba and Wold (1975). Another method is the R criterion proposed by Rice (1983, 1984). The second generation methods were proposed and studied by many researchers and considered that they are superior to the first generation methods. Park and Marron (1990) and Jones and Sheather (1991) developed plug-in method, Marron (1992) and Gasser, Kneip and Köhler (1991) extended it into the iterative plug-in method. Meanwhile, Jones, Marron and Park (1991) investigated some root n convergent methods. Then, Härdle, Hall and Marron (1992) introduced the double smoothing selection method based on the study of Müller (1985). Cao (1993) proposed the smoothed bootstrap method. Fan and Gijbels (1995) proposed a refined bandwidth selector. And Heiler and Feng (1998) improved the double smoothing method, proposed the factorized double smoothing bandwidth selector.

This section begins with a historical review of methods used in the bandwidth selection problem. Section 5.1.1 through 5.1.3 then provides a brief introduction of some common methods, R criterion, cross-validation, and double smoothing. Then followed by the plug-in method, which is the focus of this section.

5.1.1. Error Criteria

The estimation error contains the bias and variance parts, so we should find the optimal bandwidth that makes these two parts in balance. As error criterion for selecting the optimal bandwidth the MASE is to be introduced. The optimal bandwidth is the one that minimizes the MASE.

The MASE is actually the expectation of the Averaged Squared Error (ASE), which is given by

$$\mathrm{ASE}(h) = \frac{1}{n}\sum_{i=1}^{n}[\hat{m}_h(x_i) - m(x_i)]^2$$

Then MASE is derived as

$$\mathrm{MASE}(h) = \frac{1}{n}E\left\{\sum_{i=1}^{n}[\hat{m}_h(x_i) - m(x_i)]^2\right\}$$

Further, we decompose MASE into the bias and the variance part.

$$V(h) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{var}[\hat{m}(x_i)] = \frac{1}{n}\,\sigma^2\sum_{i=1}^{n}\sum_{j=1}^{n}w_{jh}(x_i)^2$$

$$B(h) = \sum_{i=1}^{n}\{E[\hat{m}_h(x_i)] - m(x_i)\}^2$$

Note that the above formal give the results in the case of i.i.d. error. If we consider the ARMA error, we have the following results, which are obtained from the sub-section 3.3.4. Here bias remains the same as it in the case of i.i.d error.

$$\mathrm{var}[\hat{m}^{(\nu)}(x)] \approx \frac{\mathbf{V}}{nh^{1+2\nu}}(R(K_2) + o(1))$$

$$h_{opt}(x) \approx \left(\frac{(k!)^2}{2k} \frac{\mathbf{V}\, R(K_2)[d-c]}{\beta_2^2 \int_c^d [m^{(k)}(x)]^2 dx} \right)^{1/2k+1} n^{-1/2k+1} = C_{opt}(x) n^{-1/2k+1}, k = 2, 4$$

Remark: By applying error criterion we can obtain bandwidth not only in the case of i.i.d error, but also ARMA error. The following three methods involving the R criterion, cross-validation, and double smoothing are only applied in the case of i.i.d error.

5.1.2. The R Criterion

Rice (1983, 1984) proposed the R criterion for bandwidth selection based on the ideas of Mallows (1973) and Craven and Wahba (1979), which is defined by

$$\hat{R}(h) = RSS(h) - \hat{\sigma}^2 + \frac{2}{n} \sum_{i=1}^{n} w_i(x_i)\, \hat{\sigma}^2 \tag{5.1}$$

where $\hat{\sigma}$ is a root n consistent estimator of the residual variance, and $w_i(x_i)$ is the weighting system for estimating $m(x_i)$.

Indeed, RSS denotes the average residual sum of squares and is given by

$$RSS(h) = \frac{1}{n} \sum_{i=1}^{n} [Y_i - \hat{m}(x_i)]^2 \tag{5.2}$$

By using $\hat{R}$, the bandwidth selected converges to the optimal bandwidth with the relative rate $n^{-1/10}$. A drawback of this method is that $\hat{R}(h)$ has large sample variation.

5.1.3. Cross-Validation

Cross-validation, also called least squared cross-validation method, is the simplest and most widely used bandwidth selection criterion. It was introduced by Clark (1975) and Wahba and

Wold (1975) in the context of smoothing splines. Discussions on the related topics can be found in the proposal of Rudemo (1982) and Bowman (1984), Härdle and Marron (1985).

The basic idea behind CV criterion is still to select bandwidth minimizing the Integrated Squared Error (ISE) defined by

$$\mathrm{ISE}(h) = \int_0^1 [\hat{m}_h(x) - m(x)]^2 f(x) dx$$

Let $\hat{h}_{cv}$ denote the cross-validation bandwidth selector, which is defined as the minimize of following formula

$$\mathrm{CV}(h) = \frac{1}{n}\sum_{i=1}^{n} [Y_i - \hat{m}_i(x_i)]^2 \qquad (5.3)$$

where $\hat{m}_i(x_i)$ is the "leave-one-out" estimator at x_i ignoring the ith observation, that is, we use data $\{(x_j, y_j), j \neq i\}$.

The least squares cross-validation bandwidth is

$$\hat{h}_{cv} = \arg\min[\mathrm{CV}(h)]$$

The great advantage of this method is that $\hat{h}_{cv}$ is the most simple bandwidth selector. Similar to the R criterion, $\hat{h}_{cv}$ also has the drawback of large sample variation. In fact, most first generation bandwidth selectors suffer from this drawback, so they might not be very useful in practice.

5.1.4. Double Smoothing

The double smoothing method was first introduced by Müller (1985) and studied latter by Härdle, Hall and Marron (1992). Remind that MASE is decomposed into variance part V and bias part B. The optimal bandwidth $\hat{h}$ is the one that minimizing $\widehat{M}(h) = \widehat{B}(h) + \widehat{V}(h)$.

Following DS criterion the variance part of MASE is estimated by

$$\widehat{V}(h) = \frac{1}{n}\,\hat{\sigma}^2 \sum_{t=1}^{n}\sum_{i=1}^{n} w_{ih}(x_t)^2 \tag{5.4}$$

where $w_{ih}(x_t)$ is the weight system with given bandwidth h.

The bias at each point x_t is estimated by means of a pilot smoothing $\hat{m}_g(x_t)$, defined as

$$\hat{b}(x_t) = \sum_{i=1}^{n} w_{ih}(x_t)\hat{m}_g(x_i) - \hat{m}_g(x_t)$$

So the bias part of MASE is estimated by

$$\widehat{B}(h) = \frac{1}{n}\sum_{t=1}^{n} \hat{b}(x_t)^2 \tag{5.5}$$

5.2. Plug-In

Another popular bandwidth selection method is the iterative plug-in method proposed by Gasser et al. (1991), which is regarded as a second generation method. Sheather (1992) and Park and Turlach (1992) compared several bandwidth selection methods and found that the plug-in method performed well and CV under smoothed. Contribution to this filed can also be found in the proposal of Herrmann et al. (1992, 1997) and Ray and Tsay (1997). The main idea of the plug-in method is to estimate the optimal bandwidths by estimating the asymptotically optimal mean integrated squared error bandwidths. The advantages of plug-in

are that, it can be easily applied to model with ARMA error, and the compute speed is quite quickly.

Recall that the asymptotically optimal bandwidth is given by

$$h_m = \left(\frac{(k!)^2}{2k} \frac{\sigma^2 \{ R(K_p) + (s-1)R(K) \}}{\int \{G^{(k)}(x)\}^2 dx \left\{ \int u^k K_p(u) du \right\}^2} \right)^{1/2k+1} n^{-1/2k+1}$$

which minimizes the dominant part of MASE. In this formula we have some unknown quantities-the variance σ^2 in the case of i.i.d, V in the case of ARMA error, and the function $I = \int \{G^{(k)}(x)\}^2 dx$. So the problem is to replace these unknown constants by some consistent estimators.

5.2.1. Estimation of Variance

In order to estimate the variance, we will introduce the difference-based methods. Rice (1984) proposed a variance estimator:

$$\hat{\sigma}^2 = \frac{1}{2(n-1)} \sum_{t=1}^{n-1} (Y_{t+1} - Y_t)^2$$

Another common used estimator was proposed by Gasser et al. (1986)

$$\tilde{\sigma}^2 = \frac{2}{3(n-2)} \sum_{t=1}^{n-2} [Y_{t+1} - \frac{1}{2}(Y_t + Y_{t+2})]^2$$

Note that $\hat{\sigma}^2$ is calculated under the i.i.d. assumption for the errors. In the following, we study the difference-based variance estimator, which is applied under ARMA assumption for the errors.

Hall et al. (1990) extended this idea to a general difference-based variance estimator with a difference sequence $D_m = \{d_0, d_1, \cdots, d_m\}$, which should satisfy $\sum_{j=0}^{m} d_j = 0$, $\sum_{j=0}^{m} d_j^2 = 1$, $m = 1, 2, \cdots$. So the variance estimator is defined by

$$\hat{\sigma}^2{}_m = (n-m)^{-1} \sum_{t=1}^{n-m} \left(\sum_{j=0}^{m} d_j Y_{t+j} \right)^2$$

$\hat{\sigma}^2{}_m$ is always a root n consistent estimator of σ^2 for any m under following assumption: G(x) is at least p+1 times differentiable and S(x) is exactly periodic with period s.

In order to estimate the variance in a seasonal time series, Heiler and Feng (1996) proposed a seasonal-difference-based variance estimator by using a seasonal difference sequence $D_{ms} = \{d_j | j = 0, 1, \cdots, m\}$, which fulfills

$$\sum_{j=0}^{m} d_j = 0, \ \sum_{j=0}^{m} d_j^2 = 1, m = 1, 2, \cdots$$

and

$$S_i = \sum_{j=0}^{m} d_j \delta_{ij} = 0, \qquad i = 0, 1, \cdots, s-1,$$

with

$$\delta_{ij} = \begin{cases} 1, & \text{if } \dfrac{j-i}{s} \text{ is an integer,} \\ 0, & \text{otherwise .} \end{cases}$$

So we can obtain the estimation of the variance

$$\hat{\sigma}^2{}_D = (n-m)^{-1} \sum_{i=1}^{n-m} \left(\sum_{j=0}^{m} d_j Y_{i+j} \right)^2 \qquad (5.6)$$

Following Hall et al. (1990) $\widehat{\sigma}^2{}_D$ is also root n consistent. For $s \geq 3$, the following seasonal difference sequence for estimating the variance is

$$D_{ms} = \frac{1}{12}\{-1, 2, -1, 0, \cdots, 0, 1, \ -2, 1\},$$

where m=s+2.

In the case of ARMA process, the acf $V = \sum_{k=-\infty}^{\infty} \gamma(k)$ is estimated by

$$\widehat{V} = \sum_{k=-k_{max}}^{k_{max}} \hat{\gamma}(k)$$

where $k_{max} = \left[n^{\frac{1}{5}}\right] + 1$. [x] means the largest integer, which is not larger than x. Hence the estimation of V is actually to obtain the estimator of acf.

$$\hat{\gamma}(k) = \frac{1}{n}\sum_{t=1}^{n-k} (x_t - \bar{x})\,(x_{t+k} - \bar{x}), \quad k = 0, 1, \ldots, k_{max} \tag{5.7}$$

For $k = -(n-1), \ldots, -1$, we define $\hat{\gamma}(-k) = \hat{\gamma}(k)$.

5.2.2. Estimation of I

As we known, to compute the plug-in bandwidth requires also the estimation of function including derivatives, which is defined by $I = \int \left\{G^{(k)}(x)\right\}^2 dx$. According to the result in the sub-section 4.1, the kth derivative of G can be estimated with a local polynomial of order p_I and a bandwidth h_I with $p_I > k$ and $p_I - k$ odd. For simplicity, we also assume that $p_I - k = 1$, so we have $\ell = p_I + 1 = k + 2$. Let $\mathbf{K}$, $\mathbf{y}$ and $\mathbf{e}_j$ are defined as before, then we obtain

$$\widehat{G}^{(k)}(t) = k!\,(\mathbf{e}_{1+k}^T, \mathbf{0}^T)(\mathbf{X}^T\mathbf{K}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{K}\mathbf{y} =: (\mathbf{w}^k)^T\mathbf{y}.$$

As a result, I can be estimated by

$$\hat{I}\left[\widehat{G}^{(k)}(x;h_I)\right] = \frac{1}{n}\sum_{i=1}^{n}\int\left\{\widehat{G}^{(k)}(x_i;h_I)\right\}^2 \qquad (5.8)$$

Next we consider some related asymptotic results of $\hat{I}$ without proof. Related results are given in Ruppert et al. (1995).

Assumption 1: $G(x_t)$ is at least ℓ times continuously differentiable.

Assumption 2: The bandwidth satisfies $h \to 0, nh^{2k+1} \to \infty, \text{as } n \to \infty$. This condition implies that h_I is of a larger order than h_m , which ensures that $\hat{I}$ is at least concistnet.

Assumption 3: S(x) is exactly periodic with period s.

Then we get

(1) $B(\hat{I}) = O({h_I}^{\ell-k}) + O[(nh_I)^{-1}{h_I}^{-2k}]$

(2) $\text{var}(\hat{I}) = O(n^{-1}) + O(n^{-2}{h_I}^{-4k-1})$

Remark: The MSE of $\hat{I}$ is dominated by the squared bias part. So by balancing the orders of the two terms on the right hand side of bias part, we can obtain the optimal bandwidth h_I that minimizing MSE, which is of the order $O(n^{-1/(\ell+k+1)})$. Note that, this is smaller than the optimal bandwidth for estimating $G^{(k)}$ itself, which is of order $O(n^{-1/(2\ell+1)})$. Moreover, the largest order of h_I should be $O(n^{-1/(4k+2)})$. A larger order of h_I will lead to the increase of the bias at the same time of keeping the variance stable.

5.2.3. The Basic Algorithm

In the following we consider only p=1, p=3 with k=2, k=4. From the viewpoint of Gasser et al. (1991), $\widehat{I_j}$, the estimate of I in the jth iteration is calculated with a bandwidth $h_{I,j}$ that can be obtained from h_{j-1} ,the bandwidth for estimating m in the (j-1)th iteration, by using an

inflation method. An inflation method is just a function $h_{I,j} = f(h_{j-1})$ so that $(h_{I,j})^{-1} = o\left[(h_{j-1})^{-1}\right]$. this means that $h_{I,j}$ will be of a larger order than h_m, if h_{j-1} is at least of order $O(h_m)$. This condition ensures that $\hat{I}$ and $\hat{h}$ will be consistent in the jth iteration.

Next we introduce two inflation methods. One is the multiplied inflation method (MIM) with the original inflation function $h_{I,j} = f(h_{j-1}) = ch_{j-1}n^{\alpha}$. It was first introduced by Gasser et al. (1991) and investigated in detail by Herrmann et al. (1992) and Ray and Tsay (1997). In this function there are some unknown quantities, c and α, which is the inflation factor with $\alpha > 0$. Indeed, the starting bandwidth h_0 is also unknown and should be set up in advance. In theory, the rate of convergence of $\hat{h}$ does not depend on c and h_0. So we will simply assume c=1. Recall that $\ell = k + 2$. Then we can obtain three reasonable choices of α for the MIM respectively.

1. $\alpha_1 = 1/(4k + 2)$ is chosen so that the order of the variance of $\hat{I}$ is minimized,
2. $\alpha_2 = 4/[(2k + 1)(2k + 5)]$ is chosen so that $\hat{G}^{(k)}$ is optimized,
3. $\alpha_3 = 2/[(2k + 1)(2k + 3)]$ is chosen so that the MSE of $\hat{I}(\hat{G}^{(k)})$ is minimized,

when convergence is reached, where $\alpha_1 > \alpha_2 > \alpha_3$.

We can easily find that α_3 is the smallest and α_3 is the optimal choice. However, this method suffers from a problem, that is, the required number of iterations by the MIM is very large, especially for $k > 2$.As pointing by Herrmann and Gasser (1994), if k=4, the iteration number is 5k+1=21 for α_1 and (k+1) (2k+1) =45 for α_3. It will be even larger in the case of long memory errors (see Ray and Tsay, 1997). So, to reduce the number of iterations required for obtaining convergence of the bandwidth, Beran (1999) proposed an exponential inflation method (EIM) with the function $h_{I,j} = f(h_{j-1}) = ch_{j-1}^{\beta}$. Beran and Feng (2002) provided some examples to show that the required number of iterations by EIM is much smaller than that by MIM. In order to get the possible choice of β, we also set c=1 and do not consider the effect of h_0. The choices of β corresponding to the $\alpha_1, \alpha_2, \alpha_3$ are:

1. $\beta_1 = 1/2$
2. $\beta_2 = (2k+1)/(2k+5)$
3. $\beta_3 = (2k+1)/(2k+3)$

where $\beta_1 < \beta_2 < \beta_3$ and β_3 is the optimal choice (Beran and Feng, 2002).

In the following we will present a basic iterative plug-in algorithm for selecting bandwidth that defined for k=2 and k=4 in time series decomposition. The algorithm is given as follows:

(1) Start with a possible bandwidth h_0

(2) For j=1,2,… set $h_{I,j} = h_{j-1}^{\beta}$ with $\beta = \beta_3 = \frac{5}{7}$ for $k = 2$ and $\beta = \beta_2 = \frac{9}{13}$ for $k = 4$. Then calculate

$$h_j = \left(\frac{(k!)^2}{2k} \frac{\sigma^2 \{R(K_p) + (s-1)R(K)\}}{\int \{G^{(k)}(x; h_{I,j})\}^2 dx \{\int u^k K_p(u) du\}^2} \right)^{1/2k+1} n^{-1/2k+1}$$

(3) Increase j by 1 and repeat step (2) until convergence is reached at some j^0 and set $\hat{h} = h_{j^0}$.

Remark 1: $\beta_1 = 1/2$ is the most stable but strongest inflation factor, which is not useful for small samples. Theoretically, β_3 is the optimal choice of β. The simulated examples and results showed that this choice works well for k=2. So we have $\beta_3 = \frac{5}{7}$ for $k = 2$. However, $\beta_3 = \frac{9}{11}$ for $k = 4$ is too close to one and the bandwidth for small samples could not be inflated correctly. Hence we choose the slightly stronger inflation factor $\beta_2 = \frac{9}{13}$. Then the variance of the estimated bandwidth with $k = 2$ and $k = 4$ is almost of the same order so that $\hat{h}$ is stable in both cases.

Remark 2: The calculation of the 4 times iterations is applied under the i.i.d. assumption for the errors. Further iterations are calculated under the ARMA assumption for the errors. The algorithm will be stopped, when $\hat{h}_j$ converges.

6. Applications

In this section, several data examples will be undertaken to investigate the methodology presented in the previous sections. The aim is to analyze the decomposition of seasonal time series by applying BV method. The trend is fitted by local polynomial, while the seasonal component is estimated by trigonometric. The model is $Y_t = G(x_t) + S(x_t) + Z_t$, Z_t is an ARMA error process as described in section 4. The bandwidth is selected according to the plug-in method. In addition, the estimation and a short-periodic forecasting of trend, seasonality, residual and the sum of them will also be considered.

6.1. Example 1

Figure 6.1 describes the results calculated for seasonal time series, where the log-scaled time series of the monthly international airline passengers from January 1949 to December 1962 is used. The order of polynomial selected is 1 and the Epanechnikov kernel is used as weight function. Furthermore, we can fit an ARMA model to the residual time series following the BIC criterion. Then, we will discuss the forecasts from January 1963 to December 1963.

Figure 6.1(a) shows the original data, which has a significant increase trend, nearly linear tendency. This is due to the large bandwidth selected in R project. From Table 6.1, the optimal bandwidth selected is 56, the total bandwidth is therefore 113. About 62% (113/12/14 years) data have been applied. Figure 6.1(b) illustrates the seasonal fluctuations, where the seasonal changing is clear and regular. As we can see, there is a peak of international airline passengers in summer and a trough in winter. It is obvious that people prefer to travel in summer than in winter. Further, comparing 1950 and 1960, the difference between the season and the off-season goes up. After adjusting the two components we can obtain the residuals as displayed in Figure 6.1 (c), which seems to be stationary. Indeed, the residuals are positive correlated from the acf of Figure 1 in Appendix. It can also be proved due to that the coefficients of the fitted model are positive and the sum of them is smaller than one.

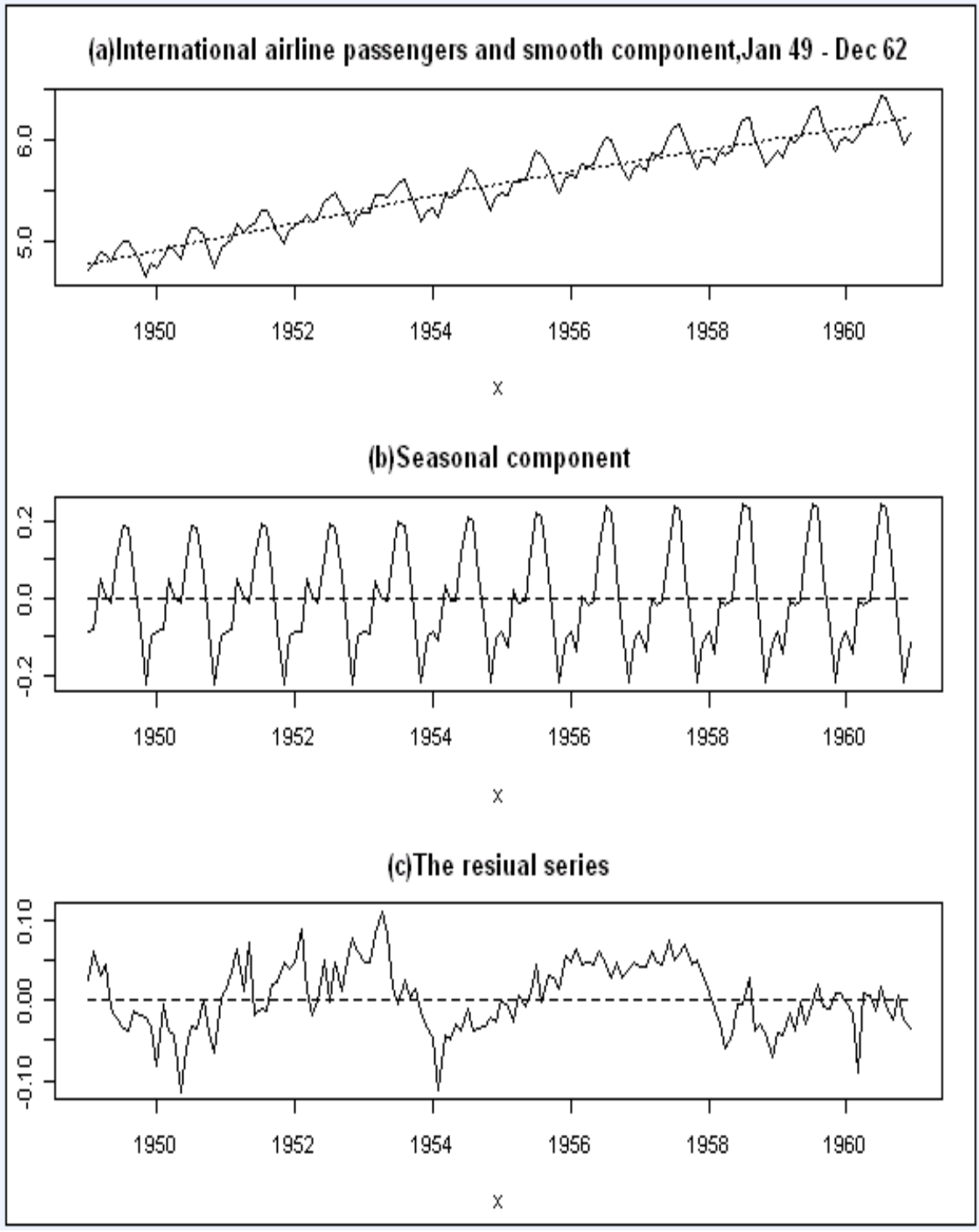


Figure 6.1: Decomposition results for time series of monthly international airline passengers from January 1949 to December 1962 with smooth trend, seasonal component and the residual.

The relevant estimated results are listed into the following table. After 10 times iteration, we can obtain the optimal bandwidth 56. Note that VarD and VarS are the variances derived from two different calculation methods, which are used to calculate the optimal bandwidth. The VarD calculated following the difference-based methods. Hence, the optimal bandwidth estimated until 4 times iteration is just the same as the start bandwidth 18, which is too small. The VarS is the variance after considering the correlation. The optimal bandwidth is therefore modified and increases until $\hat{h}_v$ stable.

n (Number of observations)	144
Number of iteration	10
$\hat{\mathbf{h}}_{\mathbf{opt}}$	56
$2\hat{\mathbf{h}}_{\mathbf{opt}} + \mathbf{1}$	113
$\hat{\mathbf{h}}_{\mathbf{4}}$	18
VarD	0.0004360177
VarS	0.008004573

Table 6.1: The estimated results in R project.

Next, we should fit an ARMA model to the residuals. First, calculate BIC for ARMA (p, q) and choose the suitable order of p, q. The BIC is given in table 6.2.

p \ q	**0**	**1**
0	-508.9748	-559.2373
1	-604.3391	-608.4435
2	***-608.7617***	-603.8375
3	-603.8309	-603.6565
4	-598.8709	-598.8729
5	-596.8050	-591.8418

Table 6.2: BIC results in R project.

It is obvious that the best model is an AR (2) model with p=2, q=0, since -608.7617 is the lowest value. Note that, if we calculate following the AIC criterion, the similar conclusion will be obtained.

Then, calculate the coefficients of ARMA model. The results are given in the following table.

	ar1	**ar2**	**intercept**
Coefficients	0.5294	0.2523	0.0047
s.e.	0.0803	0.0809	0.0104

Table 6.3: The coefficients in R project.

Hence, the selected best model is therefore an AR (2) model given by (denote the residuals by Z_t):

$$\hat{Z}_t - \hat{\mu} = 0.5294(\hat{Z}_{t-1} - \hat{\mu}) + 0.2523(\hat{Z}_{t-2} - \hat{\mu}) + \epsilon_t$$

$$\Rightarrow \hat{Z}_t - 0.0047 = 0.5294(\hat{Z}_{t-1} - 0.0047) + 0.2523(\hat{Z}_{t-2} - 0.0047) + \epsilon_t$$

In the following, we will calculate 12 point (months) predictions of trend, seasonality, residual and the sum of them. Since the interval forecasting is too complex, we only consider the point forecasting. The forecasts model is defined by

$$\hat{Y}_{n+k} = \hat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$$

It is obvious that it consist of three parts, which will be presented separately.

First, the trend component is assumed to increase according to linear extrapolation. The amount of monthly growth is defined as a constant by $\Delta G_n = \hat{G}_n - \hat{G}_{n-1}$. In this example, $\Delta G_n = 6.2179 - 6.2090 = 0.0089$, where $\hat{G}_n$ and $\hat{G}_{n-1}$ are obtained in R project.

As a result, the forecasts of trend is given by

$$\hat{G}_{n+k} = \hat{G}_n + k\,\Delta G_n$$

For k=1, $\hat{G}_{n+1} = \hat{G}_n + 0.0089 = 6.2268$.

Second, suppose that the seasonality changing is constant, because it shifts quite slowly and insignificantly. Therefore, we can apply the estimates of seasonality in the last 12 months to forecast it in the next 12 months, that is

$$\hat{S}_{n+k} = \hat{S}(n - s + k)$$

For k=1, $\hat{S}_{n+1} = \hat{S}(n - 12 + 1) = \hat{S}(133) = -0.0857.$

In addition, as we have explained in sub-section 3.5.1, the forecasts of residual is given by

$$\text{For k=1, } \hat{Z}_{n+1} - \hat{\mu} = \beta_1(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+1-p} - \hat{\mu})$$

$$\text{For k=2, } \hat{Z}_{n+2} - \hat{\mu} = \beta_1(\hat{Z}_{n+1} - \hat{\mu}) + \beta_2(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+2-p} - \hat{\mu})$$

$$\text{For k > p, } \hat{Z}_{n+k} - \hat{\mu} = \beta_1(\hat{Z}_{n+k-1} - \hat{\mu}) + \cdots + \beta_p(Z_{n+k-p} - \hat{\mu})$$

For k=1, $\hat{Z}_{n+1} - 0.0047 = 0.5294(Z_n - 0.0047) + 0.2523(Z_{n+1-2} - 0.0047)$

$$\Longrightarrow \hat{Z}_{n+1} = -0.0236$$

To calculate we need the estimated values of $Z_n = -0.0346$ and $Z_{n-1} = -0.0252$, which can be computed by R project.

For k=1, the sum of the three forecasts is

$$\hat{Y}_{n+1} = \hat{G}_{n+1} + \hat{S}_{n+1} + \hat{Z}_{n+1}$$

$$= 6.2268 - 0.0857 - 0.0236$$

$$= 6.1174.$$

As a conclusion, the forecasts of trend, seasonality, residual and the sum of them for k=1, 2… 12(i.e. from January 1963 to December 1963) are listed in the following table:

	K=1	**K=2**	**K=3**	**⋯**	**K=12**
$\widehat{\mathbf{G}}_{\mathbf{n+k}}$	6.2268	6.2357	6.2446	⋯	6.3247
$\widehat{\mathbf{S}}_{\mathbf{n+k}}$	-0.0857	-0.1428	-0.0089	⋯	-0.1149
$\widehat{\mathbf{Z}}_{\mathbf{n+k}}$	-0.0236	-0.0202	-0.0157	⋯	0.0007
$\sum$	6.1174	6.0727	6.2201	⋯	6.2106

Table 6.4: Forecasts of trend, seasonality, residual and the sum of them for one year.

Figure 6.2 shows the trend plus seasonality and forecasts for one year. Note that the above estimates and forecasts are applied after logarithmically transformation of data. The results are showed in the diagram upper. As we can see, the curve (short dashes) starting from the vertical line (n=144), which has similar movement as before, describes the forecasts for one year. The diagram below gives another expression, which extends the results to original data by exponential transformation. Essentially, it can be seen as a multiplicative model composed by exponential trend, exponential seasonality and exponential residuals.

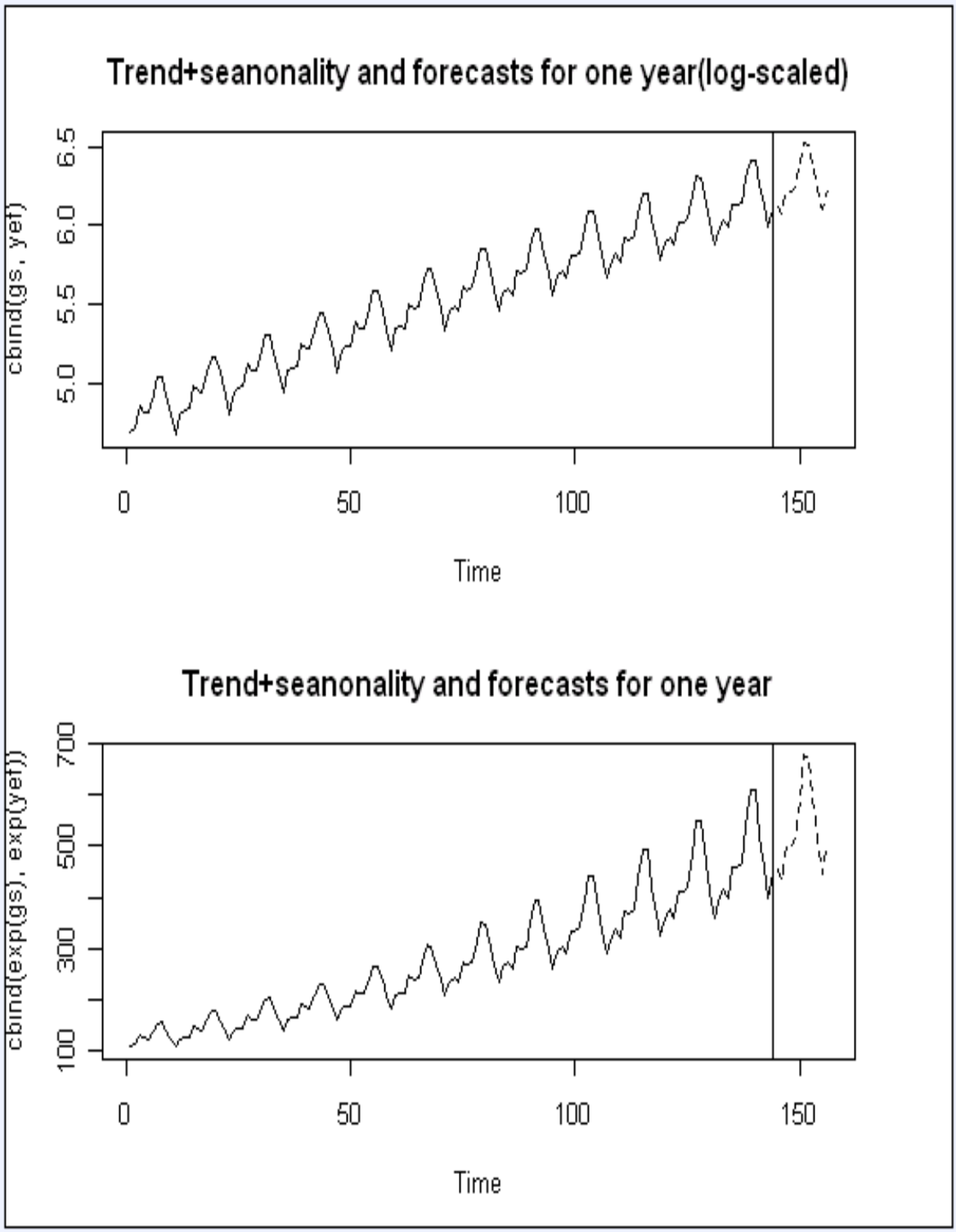


Figure 6.2: The sum of trend and seasonality with forecasts for one year (log-scaled and exponential)

6.2. Example 2

The second example is the log-scaled time series of total takings from accommodation from January 1980 to June 1995, which is also a typical example that has regular seasonal fluctuations and simple trend tendency. Like example 1, the order of polynomial selected is p=1 and the Epanechnikov kernel is used as weight function. After the estimations of trend and seasonality by locally weighted regression, and the selection of ARMA model for residual, the forecasts of trend, seasonality, residual and the sum of them from July 1995 to June 1996 will be calculated.

From Figure 6.3 (a) we can see that the data series shows an increase as the years progressed. As the economic develop quickly, the consumption level of people is also improved significantly. After 15 times iteration we get the optimal bandwidth 32, hence the total bandwidth is 65. About 34% (65/12/16 years) data have been applied. Comparing with example 1, the trend fluctuations are more complex. Specifically, we notice that the trend has two phases: first phase is a rapid growth, nearly linear tendency from January1980 to February 1992; from about February 1992 begin the second phase of a slow increase, while the residuals at this moment become negative. Figure 6.3(b) illustrates the seasonal fluctuations, which has clear seasonal changing in regularity. However, the seasonal fluctuations range is about 3% of the trend fluctuations range, which is insignificant comparing with 10% of that in example 1. As we can see, there are two peaks-summer and Christmas and in subsequent two troughs. Usually people travel in summer and in the biggest holiday- Christmas, so that the total takings from accommodation reach the peak season. Then followed by the two troughs after summer and Christmas, where the latter one changed more than the former. The residuals are displayed in Figure 6.3 (c), which seems to be stationary. From the acf of Figure 1 in Appendix we see that the residuals depend on each other positively. However, the correlation is not so strong like example 1. The related results are given in table 6.5 including number of observations, the optimal bandwidth, variance based two different calculations.

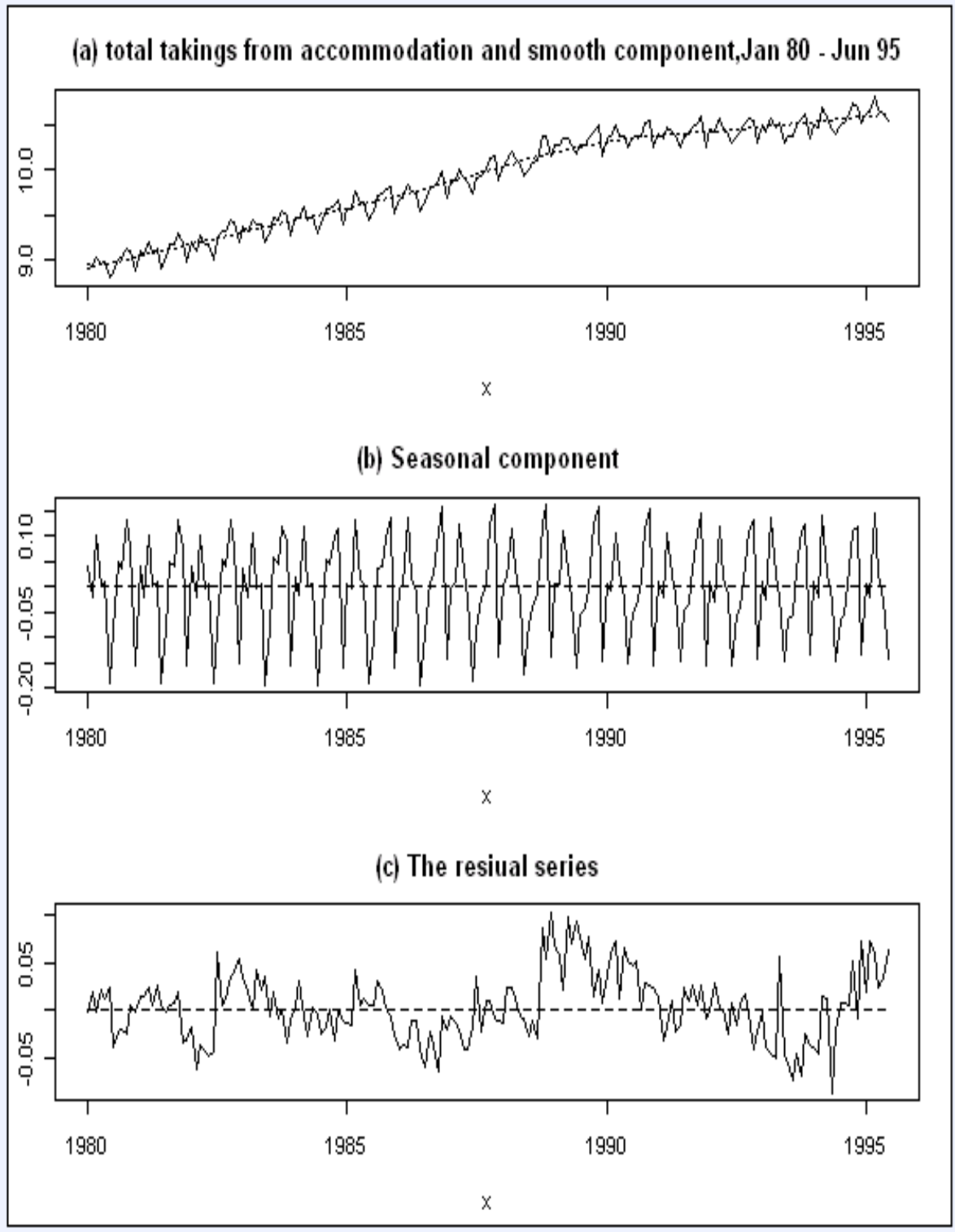


Figure 6.3: Decomposition results for time series of total takings from accommodation from January 1980 to June 1995 with smooth trend, seasonal component and the residual.

Number of observations	186
Number of iteration	15
$\hat{\mathbf{h}}_{\mathbf{opt}}$	32
$2\hat{\mathbf{h}}_{\mathbf{opt}} + \mathbf{1}$	65
$\hat{\mathbf{h}}_{\mathbf{4}}$	19
VarD	0.0008139314
VarS	0.005327541

Table 6.5: The estimated results in R project.

Next, we need to fit an ARMA model to the residual time series. Again, calculate BIC for ARMA (p, q) and select the suitable order of p, q.

p \ q	**0**	**1**
0	-710.5774	-746.2326
1	-780.5886	***-799.0312***
2	***-796.7895***	-794.5988
3	-793.0177	-789.5007
4	-792.1050	-788.3511
5	-787.9277	-783.1908

Table 6.6: BIC results in R project.

The selected best model following BIC is the ARMA (1, 1) with p=1, q=1, since -799.0312 is the lowest value. Then calculate the coefficients.

	ar1	**ma1**	**intercept**
Coefficients	0.8979	-0.5251	0.0056
s.e.	0.0435	0.0798	0.0090

Table 6.7: The coefficients in R project.

The fitted model is therefore an ARMA (1, 1) model given by:

$$\hat{Z}_t - 0.0056 = 0.8979(\hat{Z}_{t-1} - 0.0056) - 0.5251\epsilon_{t-1} + \epsilon_t$$

However, for simplicity, we only consider the forecasts of AR model, so the parameters selected is p=2, q=0 with the lowest value -796.7895. The coefficients is given below:

	ar1	**ar2**	**intercept**
Coefficients	0.3878	0.3310	0.0046
s.e.	0.0692	0.0693	0.0071

Table 6.7': The coefficients in R project.

The fitted AR (2) model is defined by

$$\hat{Z}_t - 0.0046 = 0.3878(\hat{Z}_{t-1} - 0.0046) + 0.3310(\hat{Z}_{t-2} - 0.0046) + \epsilon_t$$

Next, the 12 months forecasts for trend, seasonality, residual and the sum of them, which performs the same as the above example, will be represented. Likewise, the model is defined by

$$\hat{Y}_{n+k} = \hat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$$

Similarly, the trend component is assumed to increase in accordance with linear extrapolation. The amount of monthly growth is defined as a constant by $\Delta G_n = \hat{G}_n - \hat{G}_{n-1} = 10.6130 - 10.6080 = 0.0050$. Hence, the forecasts of trend is given by

$$\hat{G}_{n+k} = \hat{G}_n + k\,\Delta G_n$$

For k=1, $\hat{G}_{n+1} = \hat{G}_n + 0.0050 = 10.6180$.

Then, suppose that the seasonality changing is constant.

$$\hat{S}_{n+k} = \hat{S}(n - s + k)$$

For k=1, $\hat{S}_{n+1} = \hat{S}(n - 12 + 1) = \hat{S}(175) = -0.0657$.

Further, the forecasts of residual is given by

$$\text{For k=1, } \hat{Z}_{n+1} - \hat{\mu} = \beta_1(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+1-p} - \hat{\mu})$$

$$\text{For k=2, } \hat{Z}_{n+2} - \hat{\mu} = \beta_1(\hat{Z}_{n+1} - \hat{\mu}) + \beta_2(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+2-p} - \hat{\mu})$$

$$\text{For k > p, } \hat{Z}_{n+k} - \hat{\mu} = \beta_1(\hat{Z}_{n+k-1} - \hat{\mu}) + \cdots + \beta_p(Z_{n+k-p} - \hat{\mu})$$

For k=1, $\hat{Z}_{n+1} - 0.0046 = 0.3878(Z_n - 0.0046) + 0.3310(Z_{n+1-2} - 0.0046)$

$$\Rightarrow \hat{Z}_{n+1} = 0.0368$$

To calculate we obtain from R the estimated values of $Z_n = 0.0642$ and $Z_{n-1} = 0.0320$. therefore, the sum of the three forecasts for k=1is

$\hat{Y}_{n+1} = \hat{G}_{n+1} + \hat{S}_{n+1} + \hat{Z}_{n+1} = 10.6180 - 0.0657 + 0.0368 = 10.5891.$

In consequence, the detailed forecasts of trend, seasonality, residual and the sum of them for k=1, 2… 12(i.e. from July 1995 to June 1996) are listed in the following table:

	K=1	**K=2**	**K=3**	**⋯**	**K=12**
$\hat{\mathbf{G}}_{\mathbf{n+k}}$	10.6181	10.6231	10.6281	⋯	10.6734
$\hat{\mathbf{S}}_{\mathbf{n+k}}$	-0.0657	-0.0564	0.0361	⋯	-0.1429
$\hat{\mathbf{Z}}_{\mathbf{n+k}}$	0.0368	0.0368	0.0277	⋯	0.0079
$\sum$	10.5891	10.6035	10.6919	⋯	10.5384

Table 6.8: Forecasts of trend, seasonality, residual and the sum of them for one year.

The sum of trend and seasonality with forecasts for one year are illustrated in Figure 6.4. The diagram upper gives the forecasts after logarithmically transformation of data. The diagram below gives another expression, which extends the results to original data by exponential transformation. As we can see, the curve (short dashes) starting from the vertical line (n=186) describes the forecasts for one year, which has the similar movement as before.

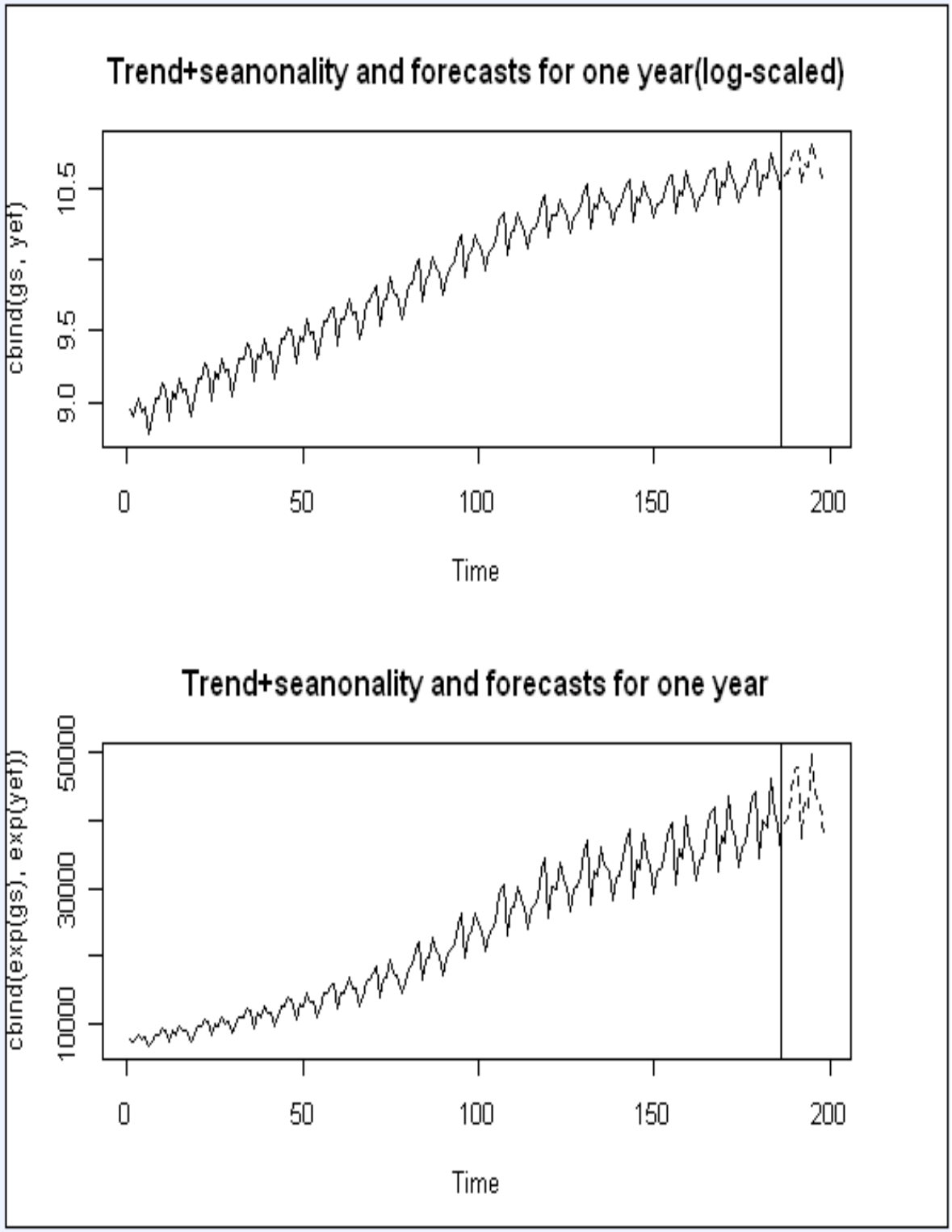


Figure 6.4: The sum of trend and seasonality with forecasts for one year (log-scaled and exponential)

6.3. Example 3

The log-scaled time series of Australia Employed Persons from February 1978 to April 1991 is chosen as the last example; this is due to the fact, that the trend fluctuations are much more complex than the first two examples. The weight function is still the Epanechnikov kernel. However, the order of polynomial selected is p=3, since the trend component is fitted better than p=1. The estimation procedure is the same as before, in the end, the forecasts of trend, seasonality, residual and the sum of them are performed from May 1991 to April 1992.

After 8 times iteration, the optimal bandwidth selected is 18; hence the total bandwidth is 37. About 22% (37/12/14 years) data have been applied. Due to the too small bandwidth, the trend and seasonality show an irregular and complex movement. This is just the reason for choosing this time series. From Figure 6.5 (a) we can see that the data series shows a slowly growth with two decreases as the years progressed. The first one is because of the 1982-1983 recession. A world wide economic crisis has taken place in many developed countries including England, Japan, Australia, and America and so on. This caused the increase of price, high inflation rate and high interest rates. So that the unemployment rate went up and the employed persons declined significantly. The second decrease of employed persons is also due to the economic crisis, which began first in the U.S.A in 1990, and then spread to Europe, Japan, Canada, Australia and other countries. The recession lasted until 1992, which is consistent with the forecasts discussed later.

The seasonal fluctuations are displayed in Figure 6.3(b), which exhibit an insignificant seasonal changing compared with the first two examples. Moreover, the seasonal fluctuations range is about 0.3% of the trend fluctuations range. Hence, the seasonality has quite small impact. The residuals are displayed in Figure 6.5 (c), which seems to be stationary. From the acf we see that the residuals have strong correlations. They depend on each other positively.

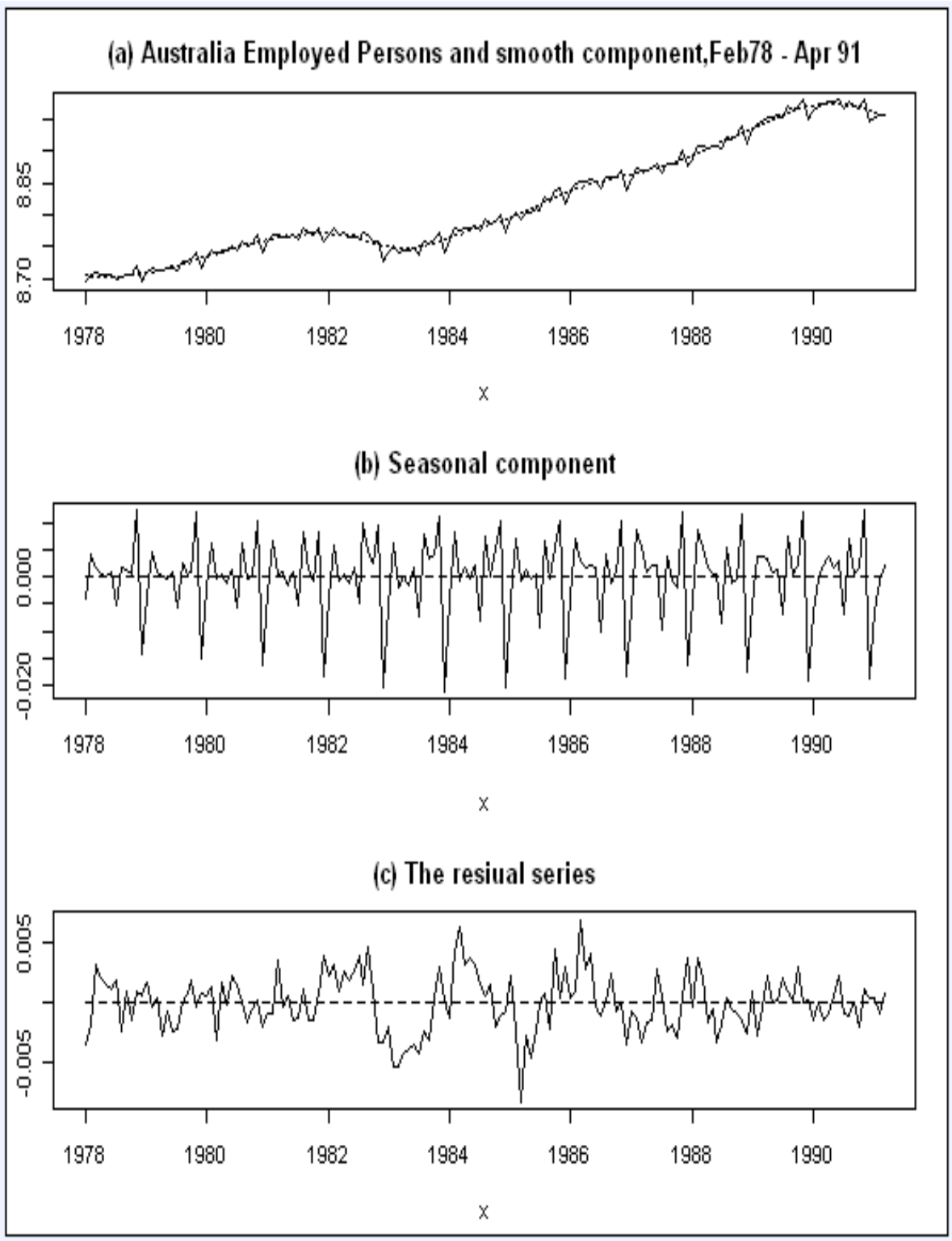


Figure 6.5: Decomposition results for time series of Australia Employed Persons from February 1978 to April 1991 with smooth trend, seasonal component and the residual.

The related results are given in table 6.9 including number of observations, the optimal bandwidth, variance based two different calculations.

Number of observations	159
Number of iteration	8
$\hat{\mathbf{h}}_{\mathbf{opt}}$	18
$\mathbf{2}\hat{\mathbf{h}}_{\mathbf{opt}} + \mathbf{1}$	37
$\hat{\mathbf{h}}_{\mathbf{4}}$	18
VarD	6.961188e-06
VarS	1.722117e-05

Table 6.9: The estimated results in R project.

p \ q	**0**	**1**
0	-1469.673	-1489.311
1	***-1501.066***	-1500.433
2	-1500.912	-1495.886
3	-1495.902	-1490.860
4	-1490.960	***-1505.272***
5	-1486.099	-1502.386

Table 6.10: BIC results in R project.

The selected model following BIC is an ARMA (4, 1) model due to the lowest value -1505.272, which is given by:

$$\hat{Z}_t = 1.2393\hat{Z}_{t-1} - 0.1299\hat{Z}_{t-2} - 0.0769\hat{Z}_{t-3} - 0.1472\hat{Z}_{t-4} + 1.0000\epsilon_{t-1} + \epsilon_t$$

With the coefficients calculated in the following table.

	ar1	**ar2**	**ar3**	**ar4**	**ma1**	**intercept**
Coefficients	1.2393	-0.1299	-0.0769	-0.1472	-1.0000	0e+00
s.e.	0.0787	0.1259	0.1270	0.0795	0.0185	1e-04

Table 6.11: The coefficients in R project.

Similarly, we only consider the forecasts of AR model, so the parameters selected is p=1, q=0 with the lowest value -1501.066. The fitted AR (1) model is defined by

$$\hat{Z}_t = 0.4544\hat{Z}_{t-1} + \epsilon_t$$

Where the coefficients are given below:

	ar1	**intercept**
Coefficients	0.4544	0e+00
s.e.	0.0708	3e-04

Table 6.11': The coefficients in R project.

The forecasts process is just the same as the above two examples, which have been clearly discussed, so we only provide the final results without detailed descriptions. The forecasts model is defined by

$$\widehat{Y}_{n+k} = \widehat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$$

Based on the same assumptions, the forecasts of trend and seasonality are given by

$$\widehat{G}_{n+k} = \widehat{G}_n + k\,\Delta G_n = \widehat{G}_n + k\left(\widehat{G}_n - \widehat{G}_{n-1}\right)$$

$$\hat{S}_{n+k} = \hat{S}(n - s + k)$$

and the forecasts of residual is given by

$$\text{For k=1, } \hat{Z}_{n+1} - \hat{\mu} = \beta_1(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+1-p} - \hat{\mu})$$

$$\text{For k > p, } \hat{Z}_{n+k} - \hat{\mu} = \beta_1(\hat{Z}_{n+k-1} - \hat{\mu}) + \cdots + \beta_p(Z_{n+k-p} - \hat{\mu})$$

For k=1, $\widehat{G}_{n+1} = \widehat{G}_n + \left(\widehat{G}_n - \widehat{G}_{n-1}\right) = 8.9524 + (8.9524 - 8.9566) = 8.9482.$

$$\hat{S}_{n+1} = \hat{S}(n - 12 + 1) = \hat{S}(148) = 0.0040.$$

$$\hat{Z}_{n+1} = 0.4544Z_n = 0.4544 \times 0.0009 = 0.0004.$$

$$\widehat{Y}_{n+1} = 8.9482 + 0.0040 + 0.0004 = 8.9526.$$

As a result, the detailed forecasts of trend, seasonality, residual and the sum of them for k=1, 2… 12(i.e. from May 1991 to April 1992) are listed in the following table:

	K=1	**K=2**	**K=3**	**⋯**	**K=6**	**⋯**	**K=12**
$\widehat{\mathbf{G}}_{\mathbf{n+k}}$	8.9482	8.9440	8.9398	⋯	8.9272	⋯	8.9020
$\widehat{\mathbf{S}}_{\mathbf{n+k}}$	0.0040	0.0016	0.0031	⋯	0.0006	⋯	0.0023
$\widehat{\mathbf{Z}}_{\mathbf{n+k}}$	0.0004	0.0002	6.1060e-05	⋯	-1.3685e-05	⋯	-2.1354e-05
$\sum$	8.9525	8.9457	8.9429	⋯	8.9277	⋯	8.9043

Table 6.12: Forecasts of trend, seasonality, residuals and the sum of them for one year.

Figure 6.6 displays the sum of trend and seasonality with forecasts for one year. The diagram upper gives the forecasts after logarithmically transformation of data. The diagram below gives another expression, which extends the results to original data by exponential transformation. As we can see, the curve (short dashes) starting from the vertical line (n=159) describes the forecasts. It shows different movements in contrast to the first two examples. Note that we cannot determine whether the prediction correct or not, since the trend fluctuations changes irregular and too complex. However, according to the actual economic status, we know that the world-wide economic recession continued until the end of 1992. In this sense, the forecasts for one year based on the model are credible and reasonable.

As a final remark, the acf of the examples are illustrated in appendix. In addition, the decomposition results of another four seasonal economic time series are exhibited.

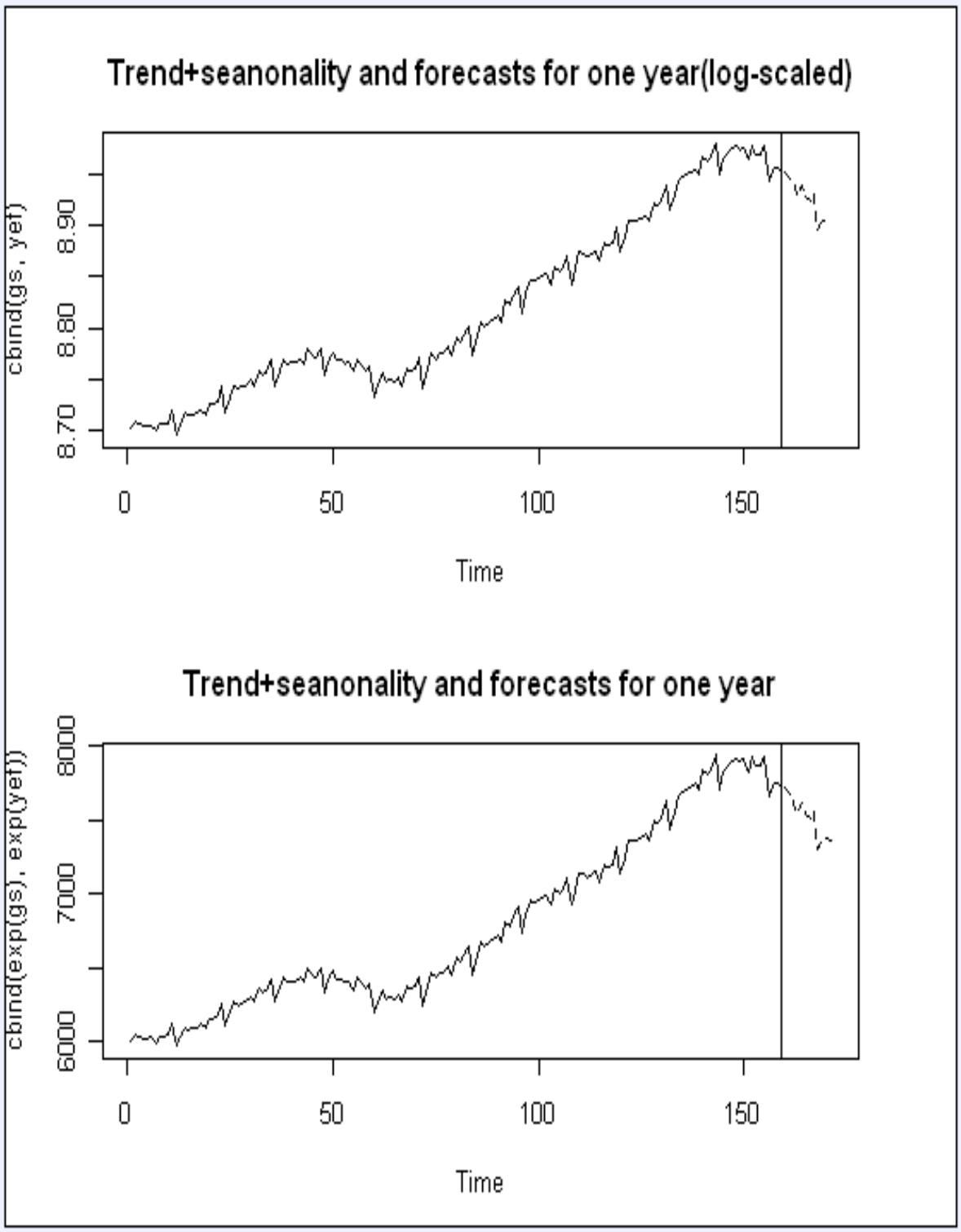


Figure 6.6: The sum of trend and seasonality with forecasts for one year (log-scaled and exponential)

7. Concluding Remarks

The major purpose of this paper is to analyze decomposition of seasonal time series based on the locally weighted regression. To deal with this, some commonly used nonparametric techniques including kernel regression and local regression, which are the basis of BV method, are detailed introduced. There are a large number of studies and proposals of many researchers have showed that nonparametric techniques can overcome the misspecification problem of parametric approaches and are much more applicable in practice. Two widely used methods-kernel regression and local regression are represented in great detail. Also, the asymptotic properties are discussed, since they are useful for applying the plug-in bandwidth selection method.

How to apply the nonparametric techniques in seasonal economic time series? We have provided a literature review and introduced some common approaches and software systems - the Census X-11 Method (1956), the SABL (1978), the Dainties (1979), the X-12 ARIMA (1988), and the SEATS (1994).compare with them, the BV (1970) method, which have been used by the German Federal Statistical Office since 1983, have some advantages: the analysis results are easy to be understood and the calculation is quite quickly; there is generally no difference between the results of direct and indirect analyses; no requirement to apply the program like user training, long experience or expert knowledge.

According to BV method, the time series are assumed to be additively composed by trend, seasonality and residual components. Further, the first two components are estimated by locally weighted regression, that is, the trend is fitted by local polynomial, while the seasonal component is estimated by trigonometric functions.

An improvement of this paper is the treatment of the residual component, which is assumed to be modeled by an ARMA process, no longer i.i.d. case. Note that the selection of suitable order of parameters follows BIC (Bayesian’s information criterion).

A crucial problem for time series decomposition is the selection of the bandwidth. There are large number methods to solve the problem. In the paper, some popular methods involving the error criterion, cross-validation, R criterion and double smoothing have been briefly introduced and judged. Compare with these methods, the plug-in method is applicable not only for the case of i.i.d error, but also ARMA error. Moreover, the compute speed is quicker and therefore the compute time is much shorter. To obtain the optimal bandwidth, the consistent estimators of variance and unknown function should be calculated in advance. Here, we have studied a seasonal-difference-based variance estimator proposed by Heiler and Feng (1996). In addition, the basic algorithm including the results following two inflation methods- the multiplied inflation method (MIM) and the exponential inflation method (EIM) have been given.

Besides, another focus of this paper is the short-term (12 months) point forecasting. The forecasts consist of three parts- trend, seasonality and residual. For simplicity, we assumed that trend increases as a constant according to linear extrapolation and the seasonal changing follows the last 12 months estimates. The forecasts of residual are based on the prediction of ARMA model. At the end of this paper, several data examples have been undertaken to investigate the methodology. Data used in this paper were mainly obtained from *Time Series Data Library* (http://robjhyndman.com/TSDL). And in order to compute and display the graphics, the best free software-R project has been applied. The data example include among consumption, gross national product, external trade and money supply. The first two data examples have a significant increase trend, nearly linear tendency. Also, the seasonality changing is clear and regular. Therefore, the 12 months point forecasts has the same movements. The third example is quite interesting, since the trend is very complex due to the two worldwide economic crises. And the seasonality has little impact that can be omitted. Hence, the forecasts based on the model and the data we have is not credible. However, if we connect it with the economic phenomena, we can find that, the possible reason of the decrease of data series is another worldwide economic crisis.

Appendix

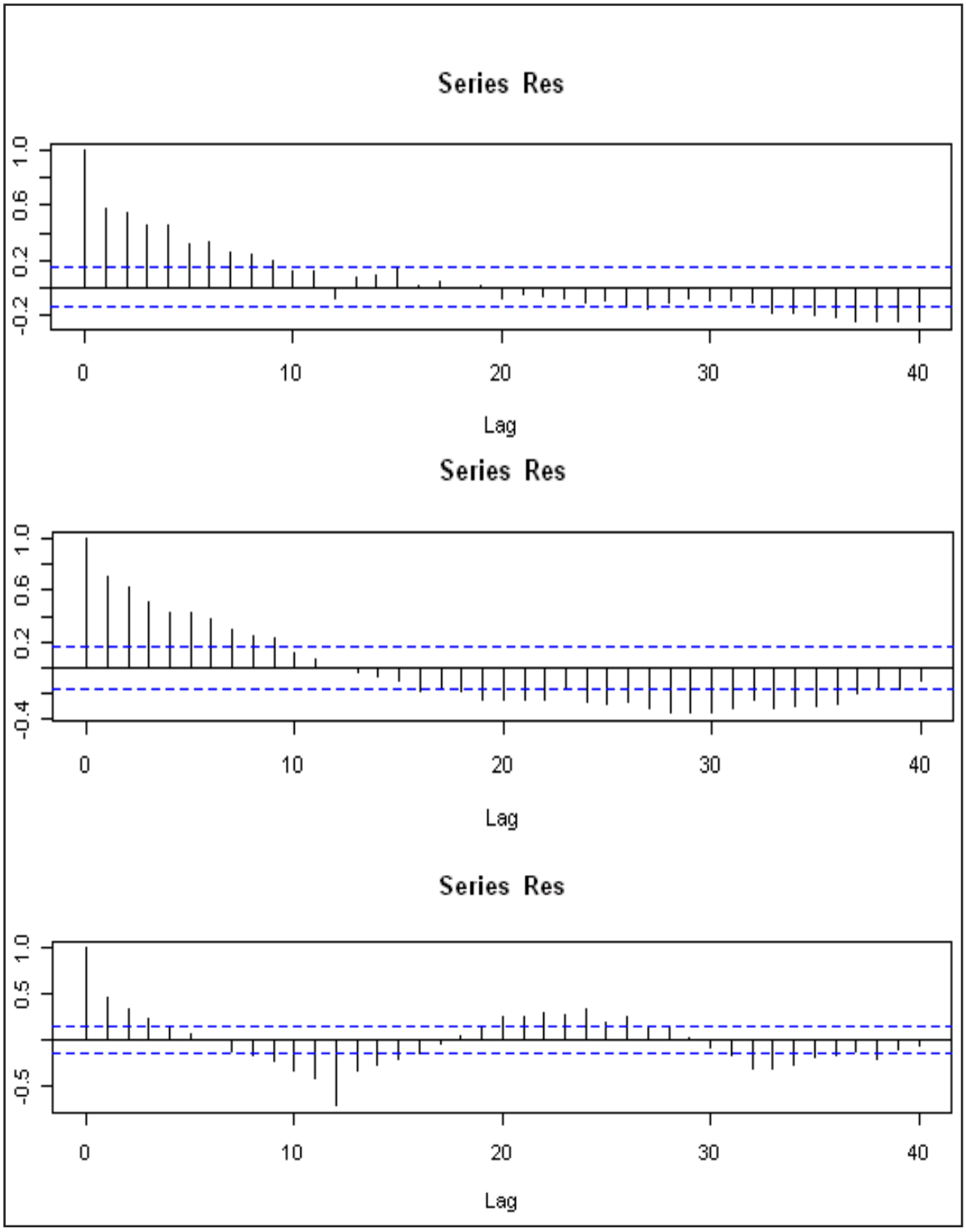


Figure 1: The acf of the residuals for example 1 (upper), example 2 (middle) and example 3 (below).

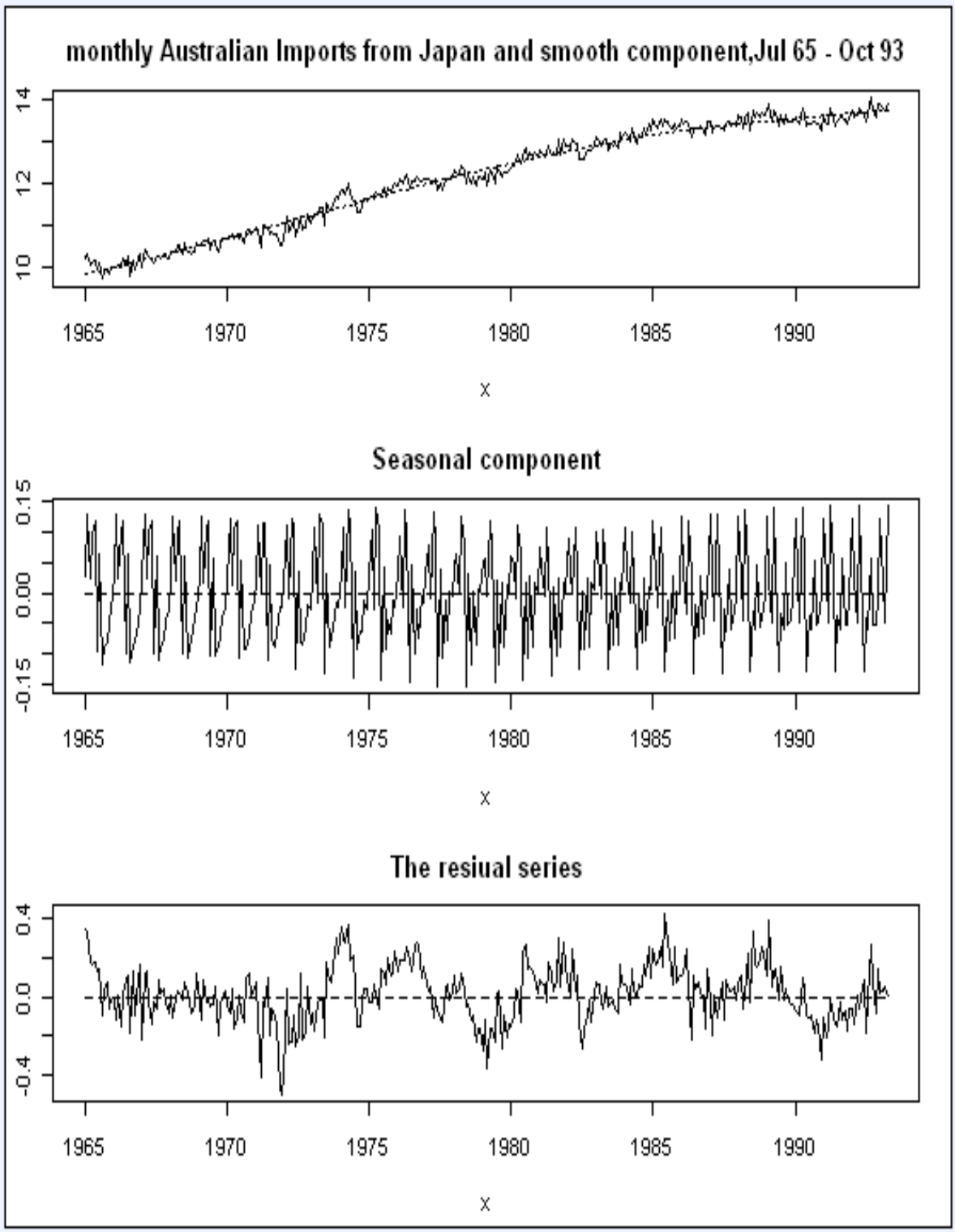


Figure 2: Decomposition results for time series of monthly Australian imports from Japan from July 1965 to October 1993 with smooth trend, seasonal component and the residual.

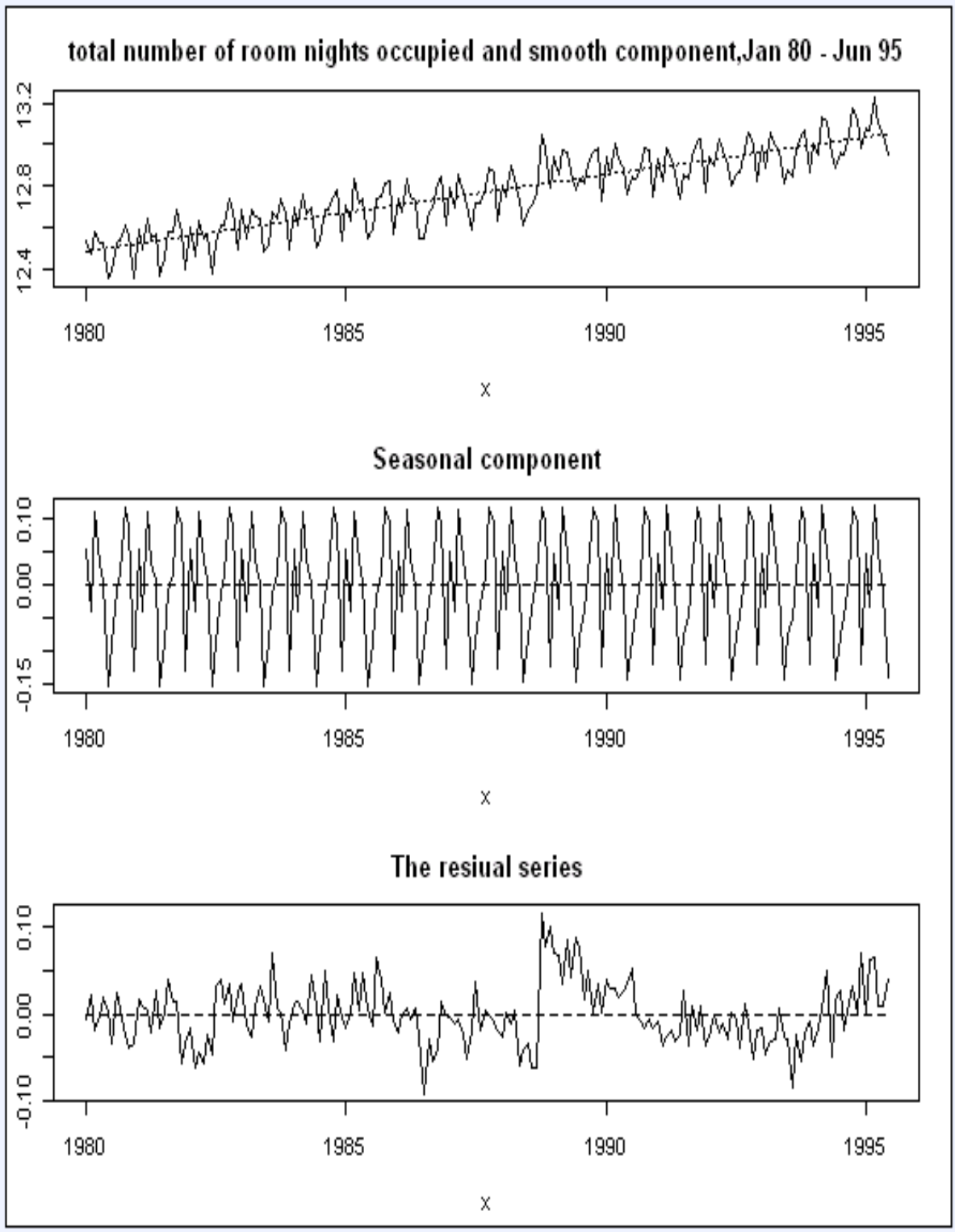


Figure 3: Decomposition results for time series of total number of room nights occupied from January 1980 to June 1995 with smooth trend, seasonal component and the residual.

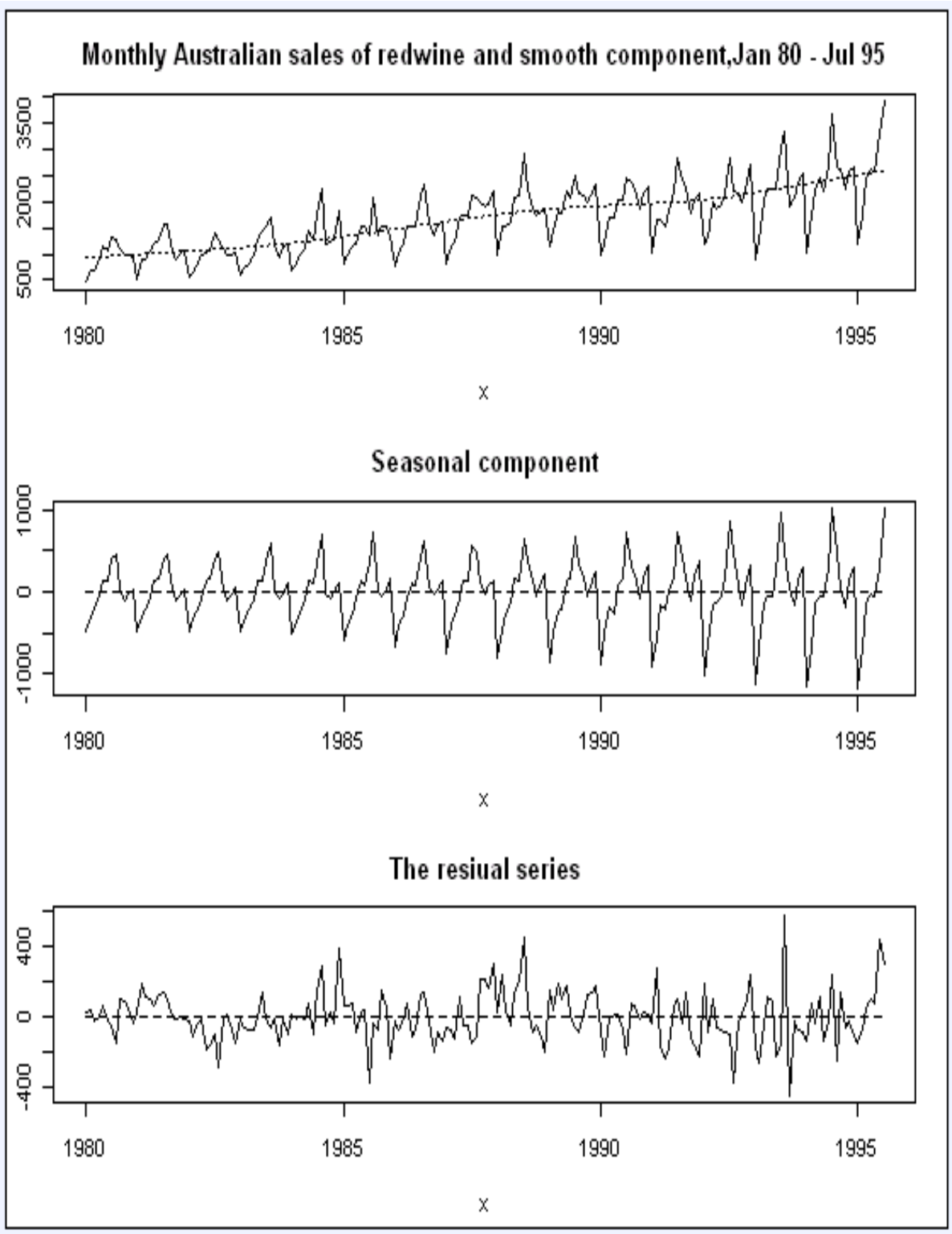


Figure 4: Decomposition results for time series of monthly Australian sales of redwine from January 1980 to July 1995 with smooth trend, seasonal component and the residual.

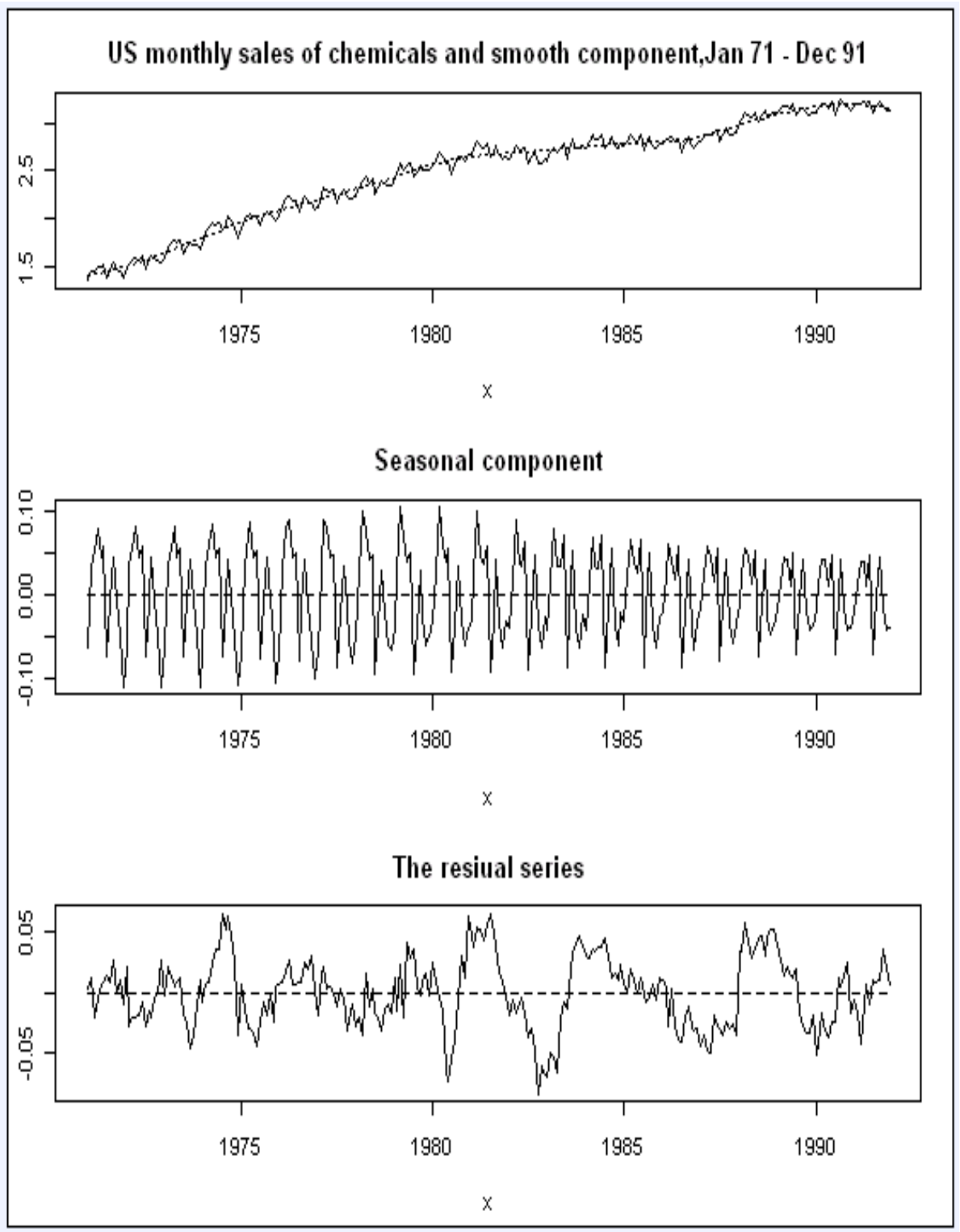


Figure 5: Decomposition results for time series of US monthly sales of chemicals from January 1971 to December 1991 with smooth trend, seasonal component and the residual.

References

[1] Barnett, W. A. (1991). *Nonparametric and semiparametric methods in Econometrics and statistics*. Cambridge Univ. Pr.

[2] Beran, J. (1999). SEMIFAR models - A semiparametric framework for modeling trends, Long range dependence and nonstationarity. Discussion paper, CoFE, 99/16, Univ.of Konstanz.

[3] Beran, J. and Feng, Y. (2001). *Local polynomial estimation with a FARIMA-GARCH error process.* Department of Mathematics and Statistics, University of Konstanz.

[4] Beveridge, S. and Nelson, C.R. (1981). A New Approach to Decomposition of Economic Time Series into Permanent and Transitory Components with Particular Attention to Measurement of the "Business Cycle," *Journal of Monetary Economics*, Vol.7, 151-74.

[5] Box, G.E.P. and Jenkins, G.M. (1976). *Time Series Analysis: Forecasting and Control*. San Francisco, Holden Day.

[6] Cleveland, R.B., Cleveland, W.S., McRAE, I.E. and Terpenning, I. (1990). STL: A Seasonal-trend Decomposition Procedure Based on LOWESS (mit Diskussion). *Journal of Offcial Statistics*, 6, 3-73.

[7] EUROSTAT (1998). Seasonal adjustment methods - A comparison. Luxembourg.

[8] Fan, J. and Yao, Q.W. (2005).*Nonlinear time series: Nonparametric and Parametric methods.* Springer.

[9] Fan, J. and Gijbels, I. (1996). *Local Polynomial Modelling and its Applications*. Chapman & Hall, London.

[10] Feng, Y. (1999). *Kernel and Locally Weighted Regression*. Verlag für Wissenschaft und Forschung, Berlin.

[11] Feng, Y. and Heiler, S. (2000). Eine robuste datengesteuerte Version des Berliner-Verfahrens (in German). *Wirtschaft und Statistik*, 10/2000, 786-795.

[12] Feng, Y. (2003). Semiparametric modelling of seasonal time series. *Bulletin of the International Statistical Institute*, Volume LX, Book 1, pp. 334-335, Berlin.

[13] Fischer, A. (1937). A Brief Note on Seasonal Variations. *Journal of Accountancy*, 64, 174.

[14] Fischer, B. (1994).Decomposition of time series –comparison between five methods of seasonal adjustment. Work report, Eurostat.

[15] Fox, J. (2000). *Nonparametric Simple Regression*. Sage Publ.

[16] Gasser, T. and Müller, H. G. (1979). Kernel estimation of regression functions, in Gasser and Rosenblatt (eds), Smoothing Techniques for Curve Estimation, *Springer Verlag*, Heidelberg.

[17] Grosser, H. (2007). Trends und Kennzahlen. *Münchner Statistik*, 21-30.

[18] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.

[19] Härdle, W. (1990). *Applied Nonparametric Regression*. Cambridge University Press, Cambridge.

[20] Härdle, W. and Chen, R. (1997). A Review of Nonparametric Time Series Analysis. *International Statistical Review*, 65, 49-72.

[21] Härdle, W. (1996). Nonparametric Time Series Model Selection. Discussion paper, Humboldt-Universitaet zu Berlin.

[22] Härdle, W. and Marron, J.S. (1985).Optimal bandwidth selection in nonparametric regression function estimation. *Ann. Statist.*, 13, 1465-1481.

[23] Hart, J. D. (1991). Kernel regression estimation with time series errors. *J.R. Statist.Soc.Ser.B*, 53,173-188.

[24] Heiler, S. (1970). Theoretische Grundlagen des 'Berliner Verfahrens'. In Sonderheft 1 zum Allg. *Statist. Archiv (edited by W. Wetzel)*, 67-93.

[25] Heiler, S. and Feng, Y. (2000). Data-driven decomposition of seasonal time series. *Journal of Statistical Planning and Inference*. 91, 351-363.

[26] Heiler, S. and Feng, Y. (1997). A Bootstrap Bandwidth Selector for Local Polynomial Fitting. *Discussion paper*, SFB178, II-344, University of Konstanz.

[27] Li, Q. and Racine, J. S. (2007). *Nonparametric Econometrics: Theory and Practice*. Princeton University Press.

[28] Lovell, M. C. (1963). Seasonal adjustment of economic time series and multiple regression analysis, *American Statistical Association Journal*, December 1963, S. 993-1010.

[29] Magauly, F.R. (1931). *The Smoothing of Time Series.*New York, National Bureau of Economic Research.

[30] Müller, H.-G. (1985).Empirical bandwidth choice for nonparametric kernel regression by means of pilot estimators. *Statist. Decisions, Suppl. Issue 2*, 193-206.

[31] Müller, H.-G. (1987). Weighted local regression and kernel methods for nonparametric curve fitting. *J. Amer.Statist. Assoc*. 82, 231-238.

[32] Nadaraya, E.A. (1964). "On Estimating Regression". Theory of Probability and its Applications 9, 141-142.

[33] Nourney, M. (1984). Seasonal adjustment by frequency determined filter procedures.*Statistical Journal of the United Nations ECE 2*, 161-168.

[34] Nullau, B., Heiler, S. u.a. (1969). Das .Berliner Verfahren". Ein Beitrag zur Zeitreihenanalyse, DIW-Beiträge zur Strukturforschung, Heft 7, Berlin.

[35] Rice, J. (1983). Methods for bandwidth choice in nonparametric kernel regression. In: Gentle, J.E. (Ed.), Computer Science and Statistics: The interface. North-Holland, Amsterdam, pp. 186-190.

[36] Rice, J. (1984). Bandwidth choice for nonparametric regression. *Ann. Statist. 12*, 1215-1230.

[37] Robinson, P.M. (1983). Nonparametric estimators for time series, *J. Time Series Anal.*, vol. 4, 185-297.

[38] Ruppert, D., Sheather, S.J. and Wand, M.P. (1995). An effective bandwidth selector for local least squares regression. *J. Amer. Statist. Assoc*., 90, 1257-1270.

[39] Siegel, S. (2001). *Nichtparametrische statistische Methoden*. 5. unveränd. Aufl. velag fuer Klotz.

[40] Simonoff, J. S. (1996). *Smoothing Methods in Statistics*. Springer, New York.

[41] Speth, H.-T. (2004). *Komponentenzerlegung und Saisonbereinigung ökonomischer Zeitreihen mit dem Verfahren BV4.1*. Statistisches Bundesamt.

[42] Statistisches Bundesamt (1999). Forschungs- und Entwicklungsplan des Statistischen Bundesamtes 1999.

[43] Time Series Data Library. (http://robjhyndman.com/TSDL)

[44] Wand, M.P. and Jones, M.C. (1995). *Kernel Smoothing*, Chapman and Hall, London.

[45] Watson, G.S. (1964). Smooth regression analysis, *Sankhya Ser. A.*

Eidesstattliche Erklärung

Hiermit versichere ich, Jing Ma, die vorliegende Arbeit selbstständig und unter ausschließlicher Verwendung der angegebenen Literatur und Hilfsmittel erstellt zu haben. Alle Stellen, die wörtlich oder sinngemäß veröffentlichtem oder unveröffentlichtem Schrifttum entnommen sind, habe ich als solche kenntlich gemacht. Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Statement of self-research thesis

The thesis submitted by Jing Ma is complete self-research work carried out under the guidance of the supervisor. Except the referenced content labeled in the paper, there is not existing any plagiarism which has been released in publications by individual or organization. Those people who make an important contribution to the research of this paper have been all mentioned obviously in the reference of the paper.

Ort	Datum	Unterschrift